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Probabilistic physics-guided transfer learning for material property prediction in extrusion deposition additive manufacturing

Akshay J. Thomas [a,b,*], Mateusz Jaszczuk [c,b], Eduardo Barocio [b], Gourab Ghosh [d], Ilias Bilionis [a,b], R. Byron Pipes [c,b]

[a] School of Mechanical Engineering, Purdue University, West Lafayette, IN, 47907, United States of America
[b] Composites Manufacturing and Simulation Center, Purdue University, West Lafayette, IN 47906, United States of America
[c] School of Aeronautics and Astronautics, Purdue University, West Lafayette, IN, 47907, United States of America
[d] Hexagon Manufacturing Intelligence, Novi, MI, 48377, United States of America

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ABSTRACT

We introduce the concept of physics-guided transfer learning to predict the thermal conductivity of an additively manufactured short-fiber reinforced polymer (SFRP) using micro-structural characteristics extracted from tensile tests. Developing composite manufacturing digital twins for SFRP composite processes like extrusion deposition additive manufacturing (EDAM) require extensive experimental material characterization. Even the same material system printed on different EDAM systems can result in significant changes to the printed micro-structure, affecting the mechanical and transport properties. This, in turn, makes characterization efforts expensive and time-consuming. Therefore, the objective of the paper is to address this experimental bottleneck and use prior information about the material manufactured in one extrusion system to predict its properties when manufactured in another system. To enable this framework, we assume that changes in properties of the same material when manufactured in different systems arise solely due to microstructural changes. To that end, we present a Bayesian framework that can transfer thermal conductivity properties across extrusion deposition additive manufacturing systems. While we discuss the transfer of thermal conductivity properties, the development is such that the framework can be used for the transfer of other properties that depend on microstructural characteristics defined in the manufacturing process. These include the coefficient of thermal expansion, viscoelastic properties, etc. The framework begins by using thermal conductivity data of the composite printed in one extrusion system to probabilistically infer the constituent thermal properties of the fiber and the polymer. Next, by conducting limited tensile tests of the same material printed in another extrusion system, we infer its orientation tensor. Finally, the inferred constituent thermal conductivity properties and the inferred orientation tensor are coupled using a micromechanics model to predict the thermal conductivity properties of the composite printed in the second extrusion system. We experimentally verify the predictions and show that our method provides a reliable framework for transferring material properties while accounting for epistemic and aleatory uncertainties.

* Corresponding author at: School of Mechanical Engineering, Purdue University, West Lafayette, IN, 47907, United States of America.
E-mail address: ajacobth@purdue.edu (A.J. Thomas).

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1. Introduction

Extrusion deposition additive manufacturing (EDAM) using short fiber-reinforced polymers has enabled the printing of large-scale structures with multiple applications, ranging from manufacturing autoclave tooling to compression molding tools [1,2]. This multi-meter scaling is possible due to the increased stiffness and lower coefficient of thermal expansion imparted to the composite by the presence of the fibers [3]. In the last decade, multiple extrusion systems have been developed to process highly filled fiber-reinforced thermoplastics. These include the Big Area Additive Manufacturing instrument developed at Oak Ridge National Lab, the Large Scale Additive Manufacturing (LSAM) system developed by Thermwood′, a medium scale system developed by Titan Robotics, and the Composites Additive Manufacturing Research Instrument (CAMRI) developed at Purdue [4]. While the manufacturing throughput of these machines varies, in working, pelletized fiber-filled polymers are extruded in a screw and deposited onto a platform using a set of machine codes. The manufacturing process is a multi-scale, multi-physics phenomenon, including anisotropic heat transfer, shrinkage, creep, and stress relaxation, polymer crystallization and melting, and fusion bonding. These mechanisms manifest as residual stresses leading to part deformation and, in some cases, failure in the form of delamination. These undesired effects can result in repeated trial and error, leading to high costs before a useful part has been manufactured. This has led to considerable efforts in developing a digital twin [5–10] of the manufacturing process to prevent such undesirable and costly outcomes. These simulation-based studies show that the heat transfer process takes a central role in the prediction of residual stresses and, consequently, residual deformation (more commonly referred to as part warpage). An important consequence of the thermal gradients developed in part is that they can dictate the extent of polymer interdiffusion between layers. The quality of the interlayer bonds governs the part strength and also plays an important role in preventing delamination [11,12] during the manufacturing process. Further, with larger bead dimensions used in the BAAM and LSAM systems, issues including delamination caused by thermal gradients across non-negligible bead dimensions are important [13,14]. The thermal gradients are important in larger beads since the large thermal mass causes higher heat retention. Therefore, a higher printing speed can cause the layer to "sag" since sufficient stiffness has not been achieved to hold subsequent layers.

Past studies have proved that heat transfer is an important aspect of the extrusion deposition process and therefore needs to be considered carefully. While numerous simulation studies have enabled a platform to understand the heat transfer process, the models still require material characterization to be performed, which can be fed as inputs to the model. Of particular interest is the thermal conductivity of the short fiber-reinforced polymer. D'Amico and Peterson [14] showed that thermal diffusivity (which in turn depends on thermal conductivity, heat capacity, and density) is an important aspect of the heat transfer process and the prediction of thermal gradients. While the heat capacity is independent of the composite microstructure (or essentially the fiber orientation and fiber length distribution), the thermal conductivity of the composite changes with fiber orientation. Under these circumstances, the same material manufactured in different processing systems possesses different properties due to the change in microstructure. The different microstructure arises due to the various flow mechanisms developed as the molten material flows through the nozzle, and is deposited onto the build platform [15]. The effect of changing processing conditions on the fiber orientation state and bead shape has been studied previously in [16]. Since different printing systems induce unique flow mechanisms (owing to changes in nozzle size, extruder geometries, etc.,) the microstructure of the printed material varies across these systems. Therefore, the main focus of this work is to establish a mathematical framework that can transfer material property information across different printing systems. We aim to address the question:"Can we use the thermal conductivity information of a material printed in one system and use it to predict the conductivity of the *same* material printed in another system without performing *any* thermal conductivity experiments".

The idea of transferring information is more commonly encountered in statistical learning research, commonly referred to as transfer learning [17]. In this method, a pre-trained model on a domain is used as the starting point to train the same model with some data in the new domain. In doing so, the model is adapted to make predictions in the new setting with significantly fewer training data points. Essentially, the model adapts to an out-of-distribution prediction here. Transfer learning presents multiple advantages when data are not abundantly available [18–20]. Note that this setting is more technically a subdivision of transfer learning called model-based transfer learning. Since the focus of this work is not to discuss the nuances of transfer learning, we point the readers to an interesting review presented in [21] for a more detailed classification and applications of transfer learning. The advantages of such transfer learning techniques have also been explored in the additive manufacturing community. Recent work on laser-based additive manufacturing presented a method to transfer displacement predictions across metal material systems [22]. In the study, the authors predicted distortion in a 316L stainless steel system, transferring the knowledge from a Ti-6Al-4V system. Menon et al. [23] studied transferring process maps from a nickel-based alloy to an iron-based alloy. Their work comprehensively studied various transfer learning settings, including data availability in the new system, the number of re-tunable parameters in the model for the target task, etc. Pandiyan et al. [24] applied the classical transfer learning approach to predict defects across different material systems. A recent work [25] presents a methodology to enhance mean-field homogenization models by training a neural network using a fast, low-fidelity model and then retraining it using a finite set of high-fidelity finite element simulations. The idea of transfer learning has also been applied to damage diagnostics in composites [26,27].

In the aforementioned transfer learning research, more often than not, a few data points in the new domain are needed to make predictions in the new domain. Let us take the example of transferring thermal conductivity information across printing systems. If one wants to transfer thermal conductivity data from the CAMRI system to the LSAM system, a few thermal conductivity experiments are still required for the material printed in LSAM as per the traditional transfer learning paradigm. However, this disregards the fact that the change in thermal conductivity is brought about by the change in microstructure when the same material is being used. To that end, we introduce *physics-guided transfer learning* that exploits known physical phenomena to enable transfer learning.

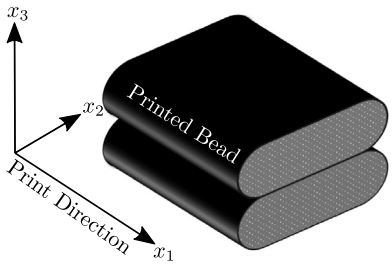

Fig. 1. Illustration of the printed bead.

Therefore, the objective of this paper is to predict the thermal conductivity of material printed in a new EDAM system by conducting tensile tests when prior information of the **same** material's thermal conductivity is available from another system. This method is demonstrated by predicting the thermal conductivity of Polyethersulfone (PESU) reinforced with 25% by weight of carbon fiber (CF) printed in the LSAM system based on prior information collected for the same material printed in the CAMRI system. The presented mathematical framework is Bayesian and therefore accounts for any uncertainties that could arise. Further, note that we present the transfer of thermal conductivity data as an example of how to build a *physics guided transfer learning* framework. Clearly, the same methodology is applicable to other properties such as the coefficient of thermal expansion or viscoelastic properties.

The remainder of the paper is organized as follows. We start by discussing the framework overview in Section 2 to guide the readers in understanding the goals and the components in the framework. We then proceed to explain each component in the framework starting with the micromechanics model and the methods used to measure the thermal conductivity of the material in Section 3. We then explain the process of Bayesian inference that accounts for uncertainties arising from epistemic uncertainties in a surrogate model in Section 4. Next, we explain the Gaussian Process surrogate that replaces the micromechanics model to keep the Bayesian inference computationally feasible in Section 5. The results are discussed in Section 6. We discuss validation experiments in Section 7 and end with conclusions in Section 8.

2. Framework overview

We provide an overview of our goal and the steps used to achieve the same. We will consider two manufacturing systems of interest, namely the LSAM and the CAMRI. Our final goal is to predict composite thermal conductivity in the LSAM system by performing only tensile tests when the thermal conductivity data of the same material in the CAMRI system is available. Throughout the remainder of the paper, we will denote the printing direction, the transverse direction to printing, and the stacking direction as 1-direction, 2-direction, and 3-direction, respectively. Fig. 1 shows an illustration of the two printed beads indicating the 1-direction using x_1, the 2-direction using x_2, and the 3-direction using x_3.

The steps involved in our framework can be visualized using the flow chart shown in Fig. 2. At the framework's core is a micromechanics model that takes the fiber orientation state of the composite, the fiber length distribution, and the thermal conductivity of the constituent polymer and fiber as inputs and predicts the thermal conductivity of the composite as the output. An orthotropic description of thermal conductivity is used to describe the printed composite, and we denote it as K_{ij}. Note that K_{ij} denotes a matrix where the entries are zero when $i \neq j$. The polymer thermal conductivity is isotropic, and the thermal conductivity of the fiber is transversely isotropic, which we denote as K_p and K_f, respectively. To describe the orientation state of the fibers, we use the second-order orientation tensor, $\mathbf{A}$, following Advani and Tucker [28]. The material is assumed to be orthotropic, and thus only the diagonal terms in the second-order orientation tensor are considered, i.e., $\mathbf{A} = \mathrm{diag}(a_{11}, a_{22}, a_{33})$ as done in [5]. Note that the term a_{ii} represents the degree of alignment of the fibers in the ith principal direction. Here, it is assumed that the principal direction coincides with the print direction.

Our mathematical framework can be summarized into the following steps:

Step 1: Infer the fiber orientation state, $\mathbf{A_1}$, of the fibers in the composite printed in the CAMRI using tensile tests in the three principal directions as performed in [29].

Step 2: Characterize the orthotropic thermal conductivity, K_{ij} of the composite printed in the CAMRI system.

Step 3: Create a Bayesian surrogate that maps the thermal conductivity of the constituent polymer and fiber, K_p and K_f, to the composite thermal conductivity, K_{ij} for the fiber orientation state $\mathbf{A_1}$ inferred in Step 1. This surrogate is created to keep the probabilistic inference in the next step computationally feasible.

Step 4: Probabilistically infer the thermal conductivity of the constituent polymer and fiber, K_p and K_f, respectively.

Step 5: Infer the fiber orientation state, $\mathbf{A_2}$, of the fibers in the composite printed in the LSAM using tensile tests in the three principal directions as performed in [29].

Step 6: Use the inferred constituent thermal conductivity properties from Step 4 and the inferred fiber orientation in Step 5 to predict the thermal conductivity in the LSAM system.

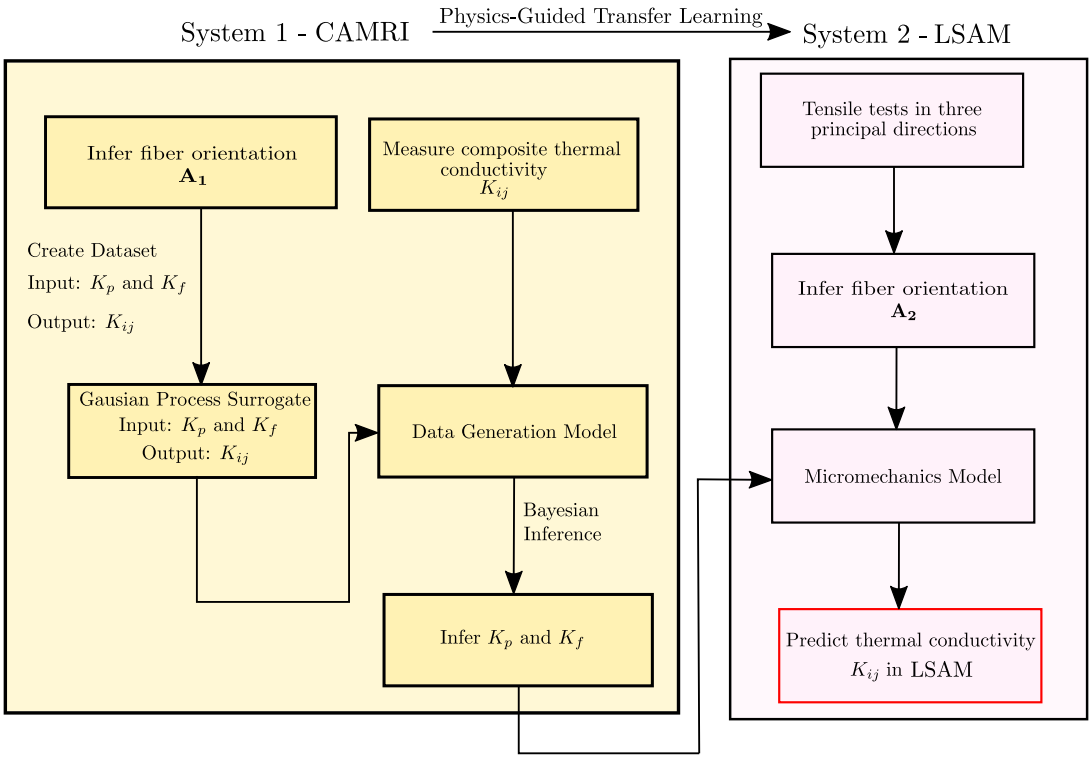

Fig. 2. Flowchart illustrating the framework.

The following sections explain each of the steps involved and how they lead to predicting the thermal conductivity of the composite in the LSAM. Note that, in the LSAM system, we predict the material's thermal conductivity just by conducting tensile tests. Tensile tests are routine experimental tests conducted to characterize the material's mechanical properties, and hence our framework presents a considerable reduction in characterization efforts to enable manufacturing simulations. Further, note that the framework is developed to infer properties in either direction. Namely, the framework could be to predict the properties in the CAMRI system when the same property has been measured in the LSAM system and vice versa.

3. Micromechanics model and experimental methods

3.1. Micromechanics model

At the core of our framework is the micromechanics model. The micromechanics model serves two purposes in our work. First, it forms a map from the constituent thermal conductivities to the thermal conductivity of the composite. This serves to create a dataset containing constituent thermal conductivities and composite thermal conductivities. This dataset is used to train a Gaussian Process that keeps the inference computationally feasible. Second, we use the micromechanics model to predict the thermal conductivity of the composite with a different microstructure once the constituent thermal conductivities of the polymer and fibers are inferred. We use a two-step homogenization method (implemented in Digimat) that uses the Mori–Tanaka model [30]. The method begins by decomposing a representative volume element into "pseudo-grains" that contain inclusions of the same orientation state and length. Next, these pseudo-grains are homogenized using a Mori–Tanaka scheme, and the readers are directed to [31] for a comprehensive derivation. Finally, the pseudo-grains are homogenized using a Voigt scheme. Note that the second step in homogenization can be carried out using various schemes, one of which is the double-inclusion model [32]. The Mori–Tanaka homogenization scheme combined with the Voigt model has been shown to provide good thermal conductivity estimates of the composite, comparable to those obtained using computational homogenization [33]. However, it is worthwhile to consider computational homogenization techniques for systems containing high-volume fractions of inclusions. In this study, fiber-to-fiber contact is not considered. Fiber-to-fiber contact provides paths of less resistance for heat transfer and therefore enhances the thermal conductivity of the composite. Since the focus of this work is to develop a framework that can transfer material properties across extrusion systems, we choose not to focus on the micromechanics model to such a level of detail. Rather, we develop a framework with rigorous statistical foundations that allow the incorporation of any suitable micromechanics model while calibrating uncertainties arising from measurement and model errors.

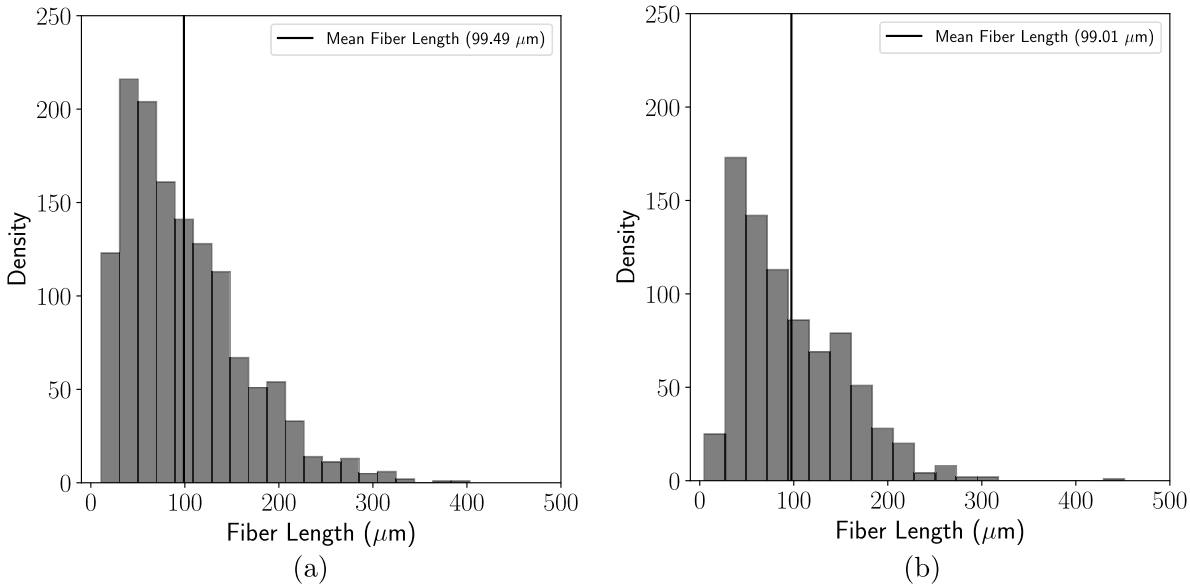

Fig. 3. Fiber length of the processed material in (a) LSAM (b) CAMRI.

Table 1
Experimental Samples Dimensions for CAMRI.

Sample type	Diameter (mm)	Thickness (mm)	Mass (mg)
1-Direction (a)	12.57	4.60	760.99
2-Direction (b)	12.60	3.23	527.51
3-Direction (c)	12.68	2.26	350.45

The inputs to the micromechanics model include the constituent polymer and fiber thermal conductivities, the fiber orientation tensor, and the fiber aspect ratio. We infer the constituent polymer and fiber thermal conductivities and this is explained in the following sections. The fiber length distribution is measured using optical microscopy after burning off the polymer and dispersing the fibers on a glass slide with silicone oil as outlined in [34]. The fiber length distributions along with the mean fiber length, of the processed materials in both the extrusion systems (LSAM and CAMRI), are shown in Fig. 3. We divide the average fiber length by the average fiber diameter (which is measured to be 5 μm) to estimate the average fiber aspect ratio. While there are some differences in the distribution, the average fiber length is almost identical. Therefore, we use an average aspect ratio of 20 in the micromechanics model for predictions in both systems. The fiber orientation state for the CAMRI is inferred using tensile tests. Our previous work has shown we can statistically infer microstructure information using tensile tests [29,35]. Following [29], we obtain the fiber orientation state of the material printed in CAMRI as $\mathbf{A}_1 = \mathrm{diag}(0.77, 0.16, 0.07)$

3.2. Experimental methods

Having discussed the micromechanics model, we now proceed to explain the experimental measurements of the composite's thermal conductivity made on the CAMRI system. The thermophysical properties required in the transient heat transfer analysis of the additive manufacturing process with composites include density, heat capacity, and thermal conductivity. While the heat capacity and the density are independent of the orientation of the fibers within a fiber-reinforced polymer, transport properties such as thermal conductivity depend on the orientation of the fibers. The thermal conductivity of an anisotropic material is defined by a second-order tensor [36]. An orthotropic description of the thermal conductivity is adopted to describe the printed material. This requires characterizing the thermal conductivity in the three principal material directions, namely in the print direction (1-direction), in-plane transverse to the print direction (2-direction), and in the layer stacking direction (3-direction). Samples in the form of disks are extracted from the beads as shown in Fig. 4. Table 1 shows the dimensions of the samples extracted. We selected those particular sample dimensions to capture the entire level of the bead in all 3 directions. The density was determined from the volume and mass of the samples.

The thermal conductivities are determined from thermal diffusivity measurements in the three principal material directions. The thermal diffusivity α was measured using the xenon flash technique according to the ASTM E1461 standard [37]. In the flash method, the front face of a small disk-shaped sample is subjected to a short laser burst, and the resulting temperature rise at the rear face of the sample is recorded and analyzed. To resolve the thermal conductivity, the density ρ and the heat capacity C_p are required ($K = \alpha C_p \rho$). The heat capacity was characterized through the DSC whereas the density was determined from measurements

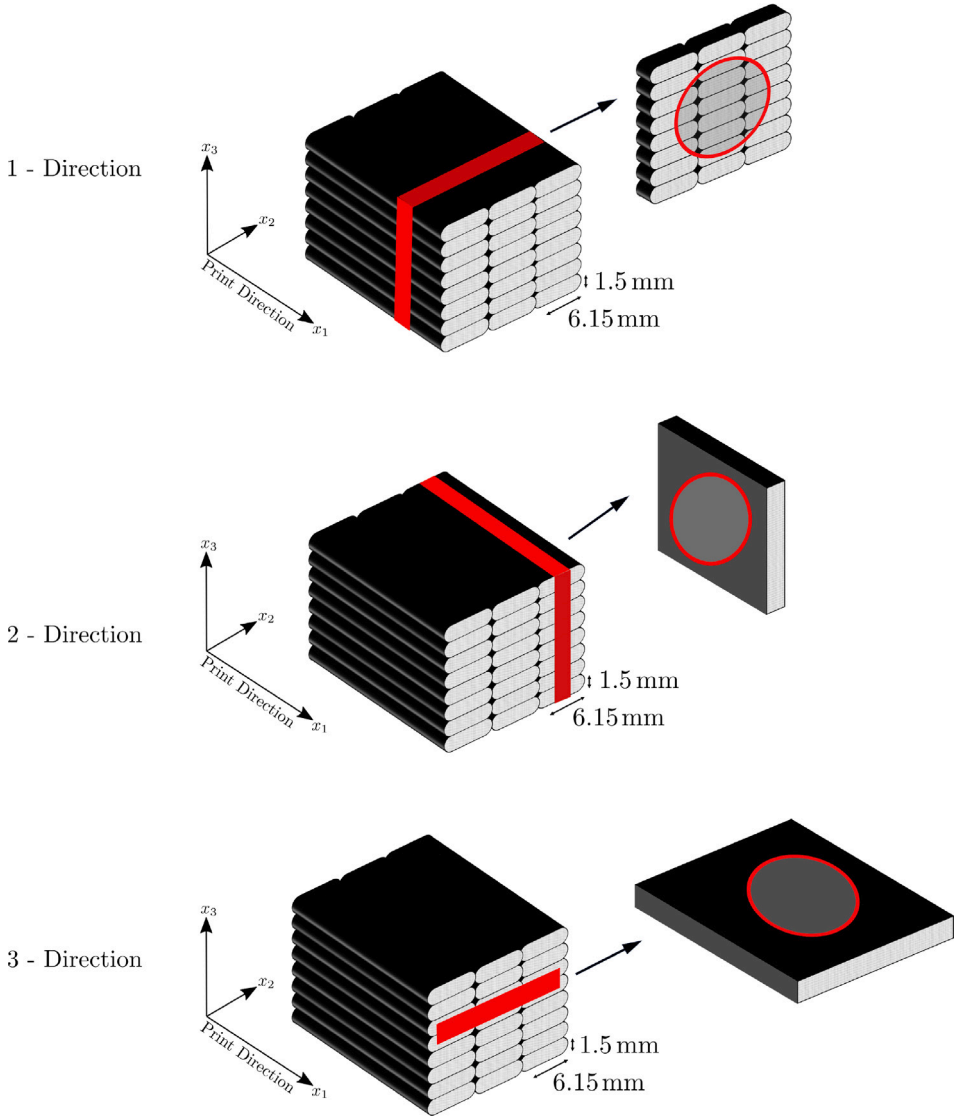

Fig. 4. Illustration showing where specimens are extracted to measure thermal conductivity.

of volume and mass at the ambient temperature. The conductivity measurements were not corrected for volumetric changes, and data was collected in a temperature range from 25 °C to 225 °C. In the study, conductivity values corresponding to temperatures above 195 °C are not included, as they lie in the glass transition region for the polymer. Fig. 5 shows the thermal conductivities, $K_{ij}, i = j = 1, 2, 3$, characterized in the three principal directions for the CF-PESU printed in the CAMRI system.

4. Bayesian inference with an uncertain micromechanics surrogate

We now discuss the steps to infer the thermal conductivity of the constituent polymer and fiber. A Bayesian inference framework that accounts for epistemic and aleatory uncertainties is used. Bayesian inference is the "gold standard" [38] of solving inverse problems. Researchers have used Bayesian inference to infer uncertain parameter values in composite materials [39,40]. Here we use the Bayesian formalism of the inverse problem to infer the constituent thermal conductivity of the polymer and the fiber. The parameters to be inferred are considered random variables whose Probability Density Function (PDF) is to be estimated. For simplicity, let $\tilde{\mathbf{x}}$ denote the parameters to be inferred. Let $f(\tilde{\mathbf{x}})$ be a model that predicts the outcome of the physical phenomenon and $\tilde{\mathbf{y}}$ be its experimentally observed value. Typically, $f(\tilde{\mathbf{x}})$ will differ from $\tilde{\mathbf{y}}$ for a variety of reasons.

This could include measurement noise, model bias, systematic bias, etc. Posing a Bayesian inference problem requires three components. First, a PDF over the unknown parameters, $\tilde{\mathbf{x}}$, is constructed. This PDF, $p(\tilde{\mathbf{x}})$, known as the *prior*, encodes the analyst's prior belief of the unknowns before any data are observed. Second, a model for the data generating process is constructed. This

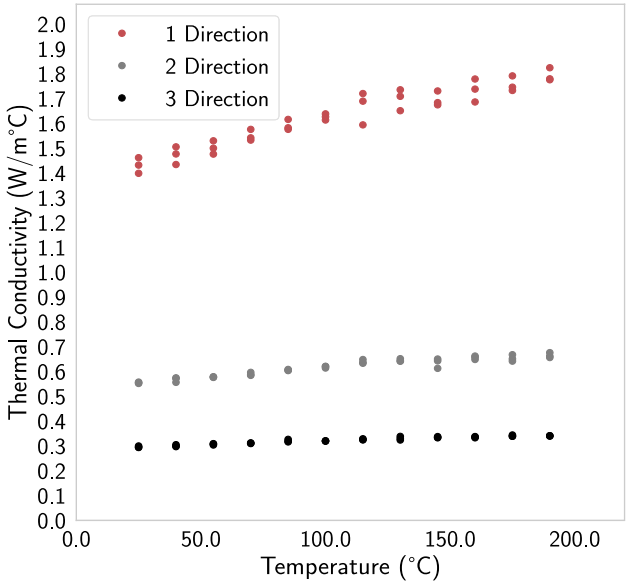

Fig. 5. Experimental composite conductivities for CAMRI.

model is known as the *likelihood* and denoted by $p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}}, f(\tilde{\mathbf{x}}))$. The likelihood quantifies the probability of measuring $\tilde{\mathbf{y}}$ given the parameters $\tilde{\mathbf{x}}$ and the model $f(\cdot)$. Third, and finally, Bayes' theorem is used to construct an updated state of knowledge about the parameters, $\tilde{\mathbf{x}}$. This is known as the posterior and can be mathematically found as:

$$p(\tilde{\mathbf{x}}|\tilde{\mathbf{y}}, f(\cdot)) = \frac{p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}}, f(\cdot))p(\tilde{\mathbf{x}})}{p(\tilde{\mathbf{y}})}, \tag{1}$$

where $p(\tilde{\mathbf{y}})$ is a normalization constant. Typically, the posterior is not analytically available, except for the case of conjugate likelihood-prior pairs [41], and therefore, sampling techniques such as the Markov Chain Monte Carlo (MCMC) [42] are generally employed. Sampling techniques require the evaluation of the forward model, and in engineering applications, this can be computationally expensive. An alternative is to replace the forward model with a low-computational cost surrogate model. Common choices of surrogates include the mean predictions of a Gaussian Process (GP) [43,44], and polynomial chaos expansion [45,46]. The quality of any surrogate depends on the amount of available forward model evaluations. To clarify, an analyst anticipates that a surrogate trained on less data is prone to be more uncertain in its prediction as opposed to the case where more data is available. Therefore, it is important to consider the uncertainties induced by limited forward evaluations. This is particularly important in multi-scale modeling of composites, where high-fidelity models are computationally expensive and only a limited number of forward evaluations are available before one exhausts time or resources. To that end, we will develop a mathematical framework that is aware of the uncertainties induced by the surrogate model. The development is such that the framework is independent of the choice of the surrogate model as long as it is Bayesian in nature.

The development of this method follows from [47]. Consider, we are given a finite number, n, of forward evaluations to create a dataset $\mathcal{D} := \{\tilde{\mathbf{x}}^i, f(\tilde{\mathbf{x}}^i)\}$, with $i = 1 \dots n$. Creating a surrogate model using $\mathcal{D}$ is essentially a regression problem, and it can be summarized using a predictive distribution. If $\tilde{\mathbf{z}}$ denotes the random variable corresponding to the output of the forward model, its predictive distribution conditioned on input is given using the sum rule:

$$p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \mathcal{D}) = \int d\boldsymbol{\theta}\ p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \theta, \mathcal{D})p(\boldsymbol{\theta}|\mathcal{D}), \tag{2}$$

where $\boldsymbol{\theta}$ contains the hyperparameters of the surrogate model, and $p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \theta, \mathcal{D})$ is the predictive distribution. The term $p(\boldsymbol{\theta}|\mathcal{D})$ is the posterior distribution of the hyperparameters. The probability density $p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \mathcal{D})$ encodes our uncertain prediction of $f(\tilde{\mathbf{x}})$ with the limited information contained in $\mathcal{D}$. We use a GP as a surrogate for the micromechanics model to keep the inference process computationally tractable. A more detailed explanation of the surrogate development is given in the next section.

We can now re-state the inverse problem as:

$$p(\tilde{\mathbf{x}}|\tilde{\mathbf{y}}, \mathcal{D}) \propto p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}}, \mathcal{D})p(\tilde{\mathbf{x}}). \tag{3}$$

Note some differences between Eq. (3) and Eq. (1). Note that in Eq. (1), the posterior was conditioned on $f(\tilde{\mathbf{x}})$. In Eq. (3), the posterior is conditioned on the dataset, $\mathcal{D}$, that is created using the forward model, $f(\cdot)$. Using the sum rule of probabilities, we can rewrite the likelihood, $p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}}, \mathcal{D})$ as:

$$p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}}, \mathcal{D}) = \int d\tilde{\mathbf{z}}\ p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}}, \tilde{\mathbf{z}})p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \mathcal{D}), \tag{4}$$

where $p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}},\tilde{\mathbf{z}})$ is the likelihood of the experimental data evaluated using $\tilde{\mathbf{z}}$ instead of $f(\tilde{\mathbf{x}})$. By substituting Eq. (2) into Eq. (4) we get the updated expression for the likelihood as:

$$p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}},\mathcal{D}) = \int d\tilde{\mathbf{z}}\ p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}},\tilde{\mathbf{z}}) \int d\boldsymbol{\theta}\ p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}},\theta,\mathcal{D})p(\theta|\mathcal{D}). \tag{5}$$

Finally, we multiply the likelihood with the prior to get the posterior as:

$$p(\tilde{\mathbf{x}}|\tilde{\mathbf{y}},\mathcal{D}) \propto \left(\int d\tilde{\mathbf{z}}\ p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}},\tilde{\mathbf{z}}) \int d\boldsymbol{\theta}\ p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}},\theta,\mathcal{D})p(\theta|\mathcal{D}) \right) p(\tilde{\mathbf{x}}). \tag{6}$$

This posterior encodes all the information we can infer about $\tilde{\mathbf{x}}$ given the experimental data $\tilde{\mathbf{y}}$ and the finite dataset $\mathcal{D}$. Further, note that the normalization constant, $p(\tilde{\mathbf{y}})$ in Eq. (6), has been dropped. Since we will be using a sampling technique, the posterior only needs to be known up to a normalization constant. However, we still have integrals over the variables θ and $\tilde{z}$. The simplest and most common approach ubiquitous in inverse problems is to replace $\tilde{z}$ with a single response surface using a set of hyper-parameters, θ^* that maximize the likelihood of the dataset used to create the surrogate. In this work, we choose to work with an intermediate setting, where we retain the predictive distribution of $\tilde{z}$. To do so, we train the Bayesian regression model using the maximum-a-posteriori (MAP) estimate to estimate θ^*. In doing so, the uncertainty induced by the predictive distribution of the surrogate, $p(\tilde{z}|\tilde{x},\theta^*)$ is retained, and we can rewrite the posterior in Eq. (6) as

$$p(\tilde{\mathbf{x}}|\tilde{\mathbf{y}},\mathcal{D}) \propto \left(\int d\tilde{\mathbf{z}}\ p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}},\tilde{\mathbf{z}})p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}},\theta^*,\mathcal{D}) \right) p(\tilde{\mathbf{x}}). \tag{7}$$

To work around the marginalization of $\tilde{\mathbf{z}}$, notice that the joint probability of $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$ given $\tilde{\mathbf{y}}, \mathcal{D}$, and θ^* is given by

$$p(\tilde{\mathbf{x}},\tilde{\mathbf{z}}|\tilde{\mathbf{y}},\mathcal{D},\theta^*) = p(\tilde{\mathbf{y}}|\tilde{\mathbf{x}},\tilde{\mathbf{z}})p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}},\theta^*,\mathcal{D})p(\tilde{\mathbf{x}}). \tag{8}$$

We can now conduct the inference in the extended space $p(\tilde{\mathbf{x}},\tilde{\mathbf{z}})$ conditioned on the experimental data $\tilde{\mathbf{y}}$, the finite dataset $\mathcal{D}$, and the MAP estimate, θ^* of the Bayesian regression model.

4.1. Likelihood function

We now explain the model for the data generating process of the experimentally observed thermal conductivity data, starting with the likelihood function. We choose to use the additive noise model where the experimentally measured thermal conductivity of the composite, K_{exp}, is related to the predicted thermal conductivity, K_{pred} as:

$$K_{\text{exp}} = K_{\text{pred}} + \text{measurement noise} \tag{9}$$

The micromechanics model is used to predict the composite thermal conductivity. However, to keep the inference computationally feasible, we replace the micromechanics model with a surrogate. If we denote the output of the surrogate using the random variable z, K_{pred} can be written as:

$$K_{\text{pred},i} = z_i(\mathbf{x}) + b_i, \text{ for } i = 1,2,3, \tag{10}$$

where $\mathbf{x}$ is the vector of polymer and fiber thermal conductivity populated as $\mathbf{x} = [K_p, K_f]$. Here, the subscript i denotes the conductivity measurement in the ith principal direction. The variable b_i accounts for a bias induced by model errors, and its probability distributed is inferred. We use the posterior distribution of a GP for z_i. For the data generating model, we assume normal measurement noise, with standard deviation τ_{exp}. Therefore, we can write the likelihood as:

$$p\left(K_{\text{exp}}|z,\mathbf{x},b,\tau_{exp}\right) = \prod_{i=1}^{3}\prod_{j=1}^{N} \mathcal{N}(K_{\text{exp},ij}|z_i(\mathbf{x}_j) + b_i, \tau_{\text{exp}}^2). \tag{11}$$

This equation represents the joint probability of observing the experimentally measured composite conductivity values given the values of constituent properties, $\mathbf{x}$, the surrogate model z, and the bias variable b_i. The double product spans each data point j for every principal direction i tested. Since we measure the thermal conductivity in three principal directions the variable i takes the value of $i = 1,2$, or 3. The variable N represents the number of data points in each principal direction. The collected dataset contains the same number of data in each principal direction.

4.2. Mathematical models for constituent properties and their priors

The thermal conductivity of the composite is dependent on temperature as is evident from the results of the experimental measurements. This implies that constituents of the composite impart temperature-dependent properties. Previous literature has shown the polymer to exhibit temperature-dependent thermal conductivities [48]. However, the thermal conductivity of the fibers is generally constant in the range of temperatures where the composite is processed for extrusion deposition additive manufacturing.

To account for the temperature-dependent thermal conductivity, we will model the thermal conductivity of the polymer, K_p, as:

$$K_p = p_1\sqrt{\frac{T}{T'}} + p_2, \tag{12}$$

where T denotes the temperature, and p_1 and p_2 are parameters to be estimated, and T' is a characteristic temperature. We take T' to be 1 °C. This modeling assumption is solely based on the observation of the experimental data for this composite. Assuming there is no fiber-to-fiber contact (this assumption is made in the micromechanics model), one expects the polymer thermal conductivity to dominate the thermal conductivity of the composites. With a different polymer system, an analyst might have to change this function. Since the focus of this work is to develop the framework for transferring information, we will not focus on specific functional forms to be used for various polymers.

Carbon fibers exhibit transversely isotropic properties for thermal conductivity. Therefore, its thermal conductivity, K_f needs two components which we will denote as K_f^l and K_f^t. The variable K_f^l denotes the thermal conductivity along the longitudinal axis of the fiber, and K_f^t denotes the component transverse to the longitudinal. We model K_f^l using two parameters l_1, and l_2 as:

$$K_f^l = l_1 T + l_2, \tag{13}$$

and K_f^t as:

$$K_f^t = \frac{K_f^l}{t}. \tag{14}$$

where t is a parameter to be estimated. The modeling choice for K_f^t is motivated from the experimental measurements made in [49], where it is observed that the transverse thermal conductivity of the AS4 and IM7 carbon fibers are approximately a constant times the average thermal conductivity in the longitudinal direction. We choose to divide by a constant in anticipation that this choice will make the inference easier, as we will estimate a value greater than one for t. To characterize the temperature-dependent thermal conductivities of the constituent properties, we need to infer the values in the list: (p_1, p_2, l_1, l_2, t).

Proceeding to place priors for the constituent properties, the priors for p_1, and p_2 are chosen as:

$$p_1 \sim \text{HalfNormal}(0.03), \tag{15}$$

and,

$$p_2 \sim \text{HalfNormal}(0.01) \tag{16}$$

respectively. Next, the priors that characterize the fibers' thermal conductivity, l_1, l_2, and t, are chosen as:

$$l_1 \sim \text{HalfNormal}(0.005), \tag{17}$$

$$l_2 \sim \text{Gamma}(10, 1), \tag{18}$$

and,

$$t \sim \text{Gamma}(5, 1). \tag{19}$$

Next, we place an exponential prior on the measurement noise standard deviation, τ_{exp}, given by:

$$\tau_{\text{exp}} \sim \text{Exponential}(\lambda_e) \tag{20}$$

Finally, we choose the priors on the bias variables in a hierarchical fashion such that:

$$\begin{aligned} b_1 &\sim \mathcal{N}(0, \tau_{b1}^2), \\ b_2 &\sim \mathcal{N}(0, \tau_{b2}^2), \\ b_3 &\sim \mathcal{N}(0, \tau_{b3}^2). \end{aligned} \tag{21}$$

The hyperpriors on τ_{b1}, τ_{b1}, and τ_{b3} in the above equations are chosen respectively as:

$$\begin{aligned} \tau_{b1} &\sim \text{HalfNormal}(0, 0.050), \\ \tau_{b2} &\sim \text{HalfNormal}(0, 0.025), \\ \tau_{b3} &\sim \text{HalfNormal}(0, 0.010). \end{aligned} \tag{22}$$

The choice of these priors induces a distribution of the thermal conductivity of the constituents as a function of temperature. Fig. 6 shows the priors induced on the constituent thermal conductivities of the constituents. Essentially, the choice of the priors induces a distribution in function space, where each sample from this distribution is a function that takes temperature as input and returns the constituent thermal conductivity. We plot the 95% credible intervals on this plot to indicate most of the mass of the distribution lies in this interval. Credible intervals (CI) are used to characterize the distribution of a random variable with point values. Formally, a credible interval of size $(1-\alpha)$ (for the parameters Θ given the data $\mathbf{X}$) in an interval (a, b) is defined as:

$$\int_a^b p(\Theta|\mathbf{X}) d\Theta = 1 - \alpha. \tag{23}$$

In short, a credible interval (a, b) contains $(1-\alpha)$% of the posterior mass. However, credible intervals are not unique since there are infinitely many intervals (a, b) that can contain $(1-\alpha)$% of the posterior mass. Therefore, we use the highest posterior density

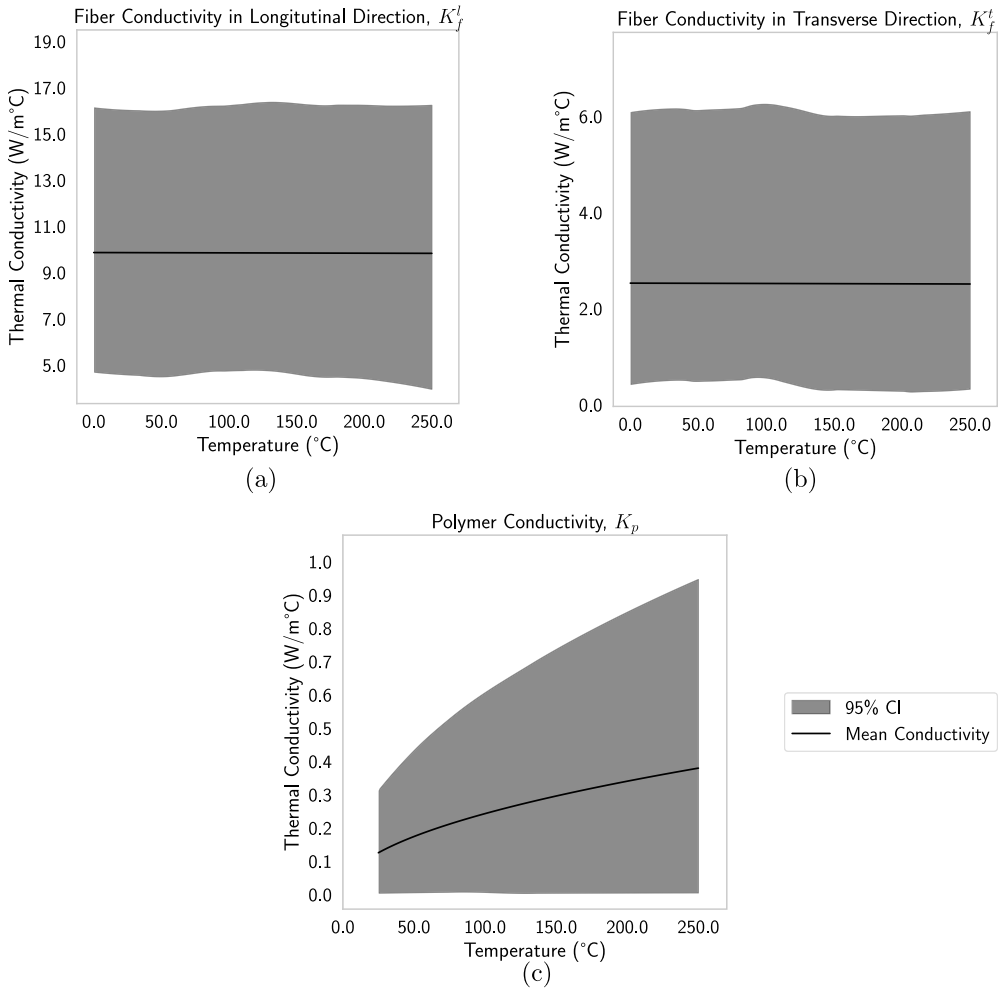

Fig. 6. Temperature dependent prior thermal conductivities (a) fiber in the longitudinal direction (b) fiber in the transverse direction, and (c) the polymer.

interval (HDI). HDI is defined as the shortest interval that contains $(1-\alpha)$% of the posterior mass. We choose a value of $\alpha = 0.05$ to plot the 95% CI. From Figs. 6 (a) and 6 (b), we see that the priors chosen induce almost no change in thermal conductivity for the fibers for the range of temperatures chosen. The values of the prior are chosen such that we enforce (weakly) the prior information that fibers remain inert in the range of processing temperatures. From Fig. 6(c) we see that the polymer thermal conductivity is a non-linear function of temperature. This choice is motivated by the observation of the experimental data and that the temperature dependence of the composite thermal conductivity is highly dependent on the polymer.

5. Gaussian process surrogate

We choose to work with sampling techniques to characterize the posterior of the parameters in the data generating model. In particular, we use the Markov Chain Monte Carlo (MCMC) [42] method to infer the parameters used in the data generating model. To ensure the convergence of the MCMC chain to characterize the posterior of the parameters, we need to draw samples in the order of 10^4–10^6 [50]. The inference process can become computationally infeasible if it relies on the evaluation of a forward model that takes even just 1s to evaluate. Therefore, we use a GP surrogate to replace the micromechanics model to avoid this computational bottleneck. In this study, we use the entire posterior of GP conditioned on a finite number of model evaluations as the micromechanics surrogate.

A GP is a collection of random variables such that a subset of this collection has a joint Gaussian distribution [41]. Consider a dataset, $\mathcal{D} = \{(\mathbf{x}^m, g^m)\colon m = 1 \dots M\}$. To clarify, $\mathcal{D}$ is a dataset of M- input–output pairs. Here, $\mathbf{x} = (K_p, K_f^l, K_f^t)$ and $\mathbf{g} = (K_1, K_2, K_3)$. The variable K_i represents the thermal conductivity of the polymer in the principal direction i. Note that, $\mathbf{g}$ has a dimension of 3 and we use the notation $g_i = f_i(\mathbf{x})$ to refer to each of the composite conductivity values. We model each $f_i(\mathbf{x})$ using a GP, defined by a mean function and a covariance function:

$$f_i(\mathbf{x})|\theta_i \sim \mathrm{GP}(0, k(\mathbf{x}, \mathbf{x}'; \theta_i)), \tag{24}$$

where θ_i are the hyperparameters of the covariance function. Our prior beliefs about the function are encoded using the mean, and covariance functions, $k(\mathbf{x}, \mathbf{x}'; \boldsymbol{\theta})$. In this study, we choose a zero-mean function, which implies that we do not enforce any prior belief on the function mean value, and the squared exponential covariance function is defined as:

$$k(\mathbf{x}^m, \mathbf{x}^n; \theta_i) = s^2 \exp\left(-\frac{1}{2}(\mathbf{x}^m - \mathbf{x}^n)^T \Lambda^{-1}(\mathbf{x}^m - \mathbf{x}^n)\right), \tag{25}$$

where $\Lambda = \mathrm{diag}(c_1^2, c_2^2, c_3^2)$ and the parameters c_1, c_2, and c_3, are called the characteristic length scales. The squared exponential covariance function encodes the belief that the model output is infinitely differentiable with respect to the input parameters. The choice of this kernel is made as we expect the thermal conductivity of the polymer to be a smooth function of the thermal conductivity of its constituents and since we do not anticipate any discontinuities. The hyperparameters in this model are $\boldsymbol{\theta} = (s, c_1, c_2, c_3)$. The number of lengthscales corresponds to the input dimension. The parameter, s, is called the signal strength. The number of lengthscales corresponds to the input dimension. The characteristic lengthscales can be informally thought of as the distance that needs to be moved in parameter space before a significant change in the function is observed. As a result, a larger length scale in a particular parameter space implies that the function output is less sensitive to changes in that particular parameter. Finding the optimal values of θ that best represents that data is called the Gaussian Process Regression (GPR).

Denoting the given input dataset as $\mathcal{D}_X$ the prior GP (using the zero mean function and covariance function) on the function response is:

$$\mathbf{f}_i | \mathcal{D}_X, \theta_i \sim \mathcal{N}(\mathbf{0}, \mathbf{K}) \tag{26}$$

where, $\mathbf{K}$ represents the covariance matrix with an entry at (m, n) defined as $k(\mathbf{x}^m, \mathbf{x}^n; \boldsymbol{\theta})$, and $\mathbf{f}_i$ represents the vector of function values populated as $\mathbf{f}_i = (f_i(x^{(1)})), f_i(x^{(2)}), \dots f_i(x^{(M)}))$. Note that, Eq. (26) represents a joint probability of the function responses conditioned on the inputs before observing any data.

A Gaussian noise can be added to each function output to obtain noisy outputs (since datasets are often contaminated with noise), h_i, defined as:

$$h_i^m = f_i^m(\mathbf{x}^m) + \epsilon_i^m, \tag{27}$$

where $m = 1 \dots M$. The variable ϵ_i represents a Gaussian noise vector with zero mean and standard deviation of $\tau_{\mathrm{GP},i}$, i.e., $\epsilon_i \sim \mathcal{N}(0, \tau_{\mathrm{GP},i}^2)$. Here, we use the GP to build a surrogate for simulation outputs, which implies that the use of the noise vector is not necessary. However, we include it in the training since it aids numerical conditioning during the factorization of the covariance matrix [51]. We set the value to be $1e-6$.

Introducing a set of test inputs, denoted by $\mathcal{D}_X^*$, and invoking that any finite subset of random variables in a GP are jointly Gaussian, the joint distribution of the observed noisy target and the function evaluations at the target inputs can be written as:

$$\begin{bmatrix} \mathbf{h}_i \\ \mathbf{f}_i^* \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K}(\mathcal{D}_X, \mathcal{D}_X) + \tau_{\mathrm{GP},i}^2 \mathbf{I} & \mathbf{K}(\mathcal{D}_X, \mathcal{D}_X^*) \\ \mathbf{K}(\mathcal{D}_X^*, \mathcal{D}_X) & \mathbf{K}(\mathcal{D}_X^*, \mathcal{D}_X^*) \end{bmatrix}\right), \tag{28}$$

where $\mathbf{h}_i$ represents the vector of noisy function outputs evaluated at $\mathcal{D}_X$, $\mathbf{f}_i^*$ represents the vector of function outputs evaluated at the test inputs, $\mathcal{D}_X^*$, and $\mathbf{K}(\mathcal{D}_X, \mathcal{D}_X^*)$ denotes the $M \times M_*$ (where M denotes the size of the training set and M_* denotes the size of the test set)matrix of the covariance values evaluated at all pairs of the training and test data points. Baye's rule is used to evaluate the posterior distribution of the GP, from which predictive mean and variance at a single test point, $\mathbf{x}^*$ can be written as:

$$\mu_i(\mathbf{x}^*; \theta_i, \tau_{\mathrm{GP},i}) = \mathbf{k}(\mathbf{x}^*, \mathcal{D}_X; \theta_i)(\mathbf{K}(\mathcal{D}_X, \mathcal{D}_X) + \tau_{\mathrm{GP},i}^2 \mathbf{I})^{-1} \mathbf{h}_i, \tag{29}$$

and

$$\tau_i^2(\mathbf{x}^*; \theta_i, \tau_{\mathrm{GP},i}) = k(\mathbf{x}^*, \mathbf{x}^*; \theta_i) - \mathbf{k}(\mathbf{x}^*, \mathcal{D}_X; \theta_i)\mathbf{K}(\mathcal{D}_X, \mathcal{D}_X) + \tau_{\mathrm{GP},i}^2 \mathbf{I})^{-1} \mathbf{k}(\mathcal{D}_X, \mathbf{x}^*; \theta_i) \tag{30}$$

respectively, where $\mathbf{I}$ is the identify matrix, and $\mathbf{k}(\mathbf{x}^*, \mathcal{D}_X; \theta_i)$ is a vector of covariance values between the test point $\mathbf{x}^*$ and the input dataset, $\mathcal{D}_X$, populated as

$$\mathbf{k}(\mathbf{x}^*, \mathcal{D}_X; \theta_i) = (k(\mathbf{x}^*, \mathbf{x}^{(1)}; \theta_i), k(\mathbf{x}^*, \mathbf{x}^{(2)}; \theta_i) \dots k(\mathbf{x}^*, \mathbf{x}^{(M)}; \theta_i)).$$

We have now set up everything required to perform GPR. To do so, we maximize the log-likelihood of output data in the training data set. The optimal values of the hyperparameters, which we denote as θ^* and $\tau_{\mathrm{GP},i}^*$, by solving an optimization problem defined as [52]:

$$\theta_i^*, \tau_{\mathrm{GP},i}^* = \mathrm{argmax} \;\; L_{\mathrm{GP}}(\theta_i, \tau_{\mathrm{GP},i}; \mathcal{D}_X, \mathbf{h}_i), \tag{31}$$

where L_{GP} denotes the likelihood of the training data. We implement the GPR using the Python package pymc3 [53] and use the Broyden Fletcher Goldfarb Shanno (BFGS) algorithm for the hyperparameter optimization. It is well known that this optimization is non-convex and there is the possibility of getting stuck at a bad local minima [54]. To ensure that we avoid bad local minima, we restart the algorithm at different starting points and pick the one with the largest log-likelihood. In the training process, it is observed that re-starting the algorithm at various points had no effect on the results.

In our earlier development, we discussed the effect of surrogate uncertainty in the inference process. To elucidate the effect of the surrogate uncertainty in the inference process, we consider two data sets to train the GP model. The first data set, which we

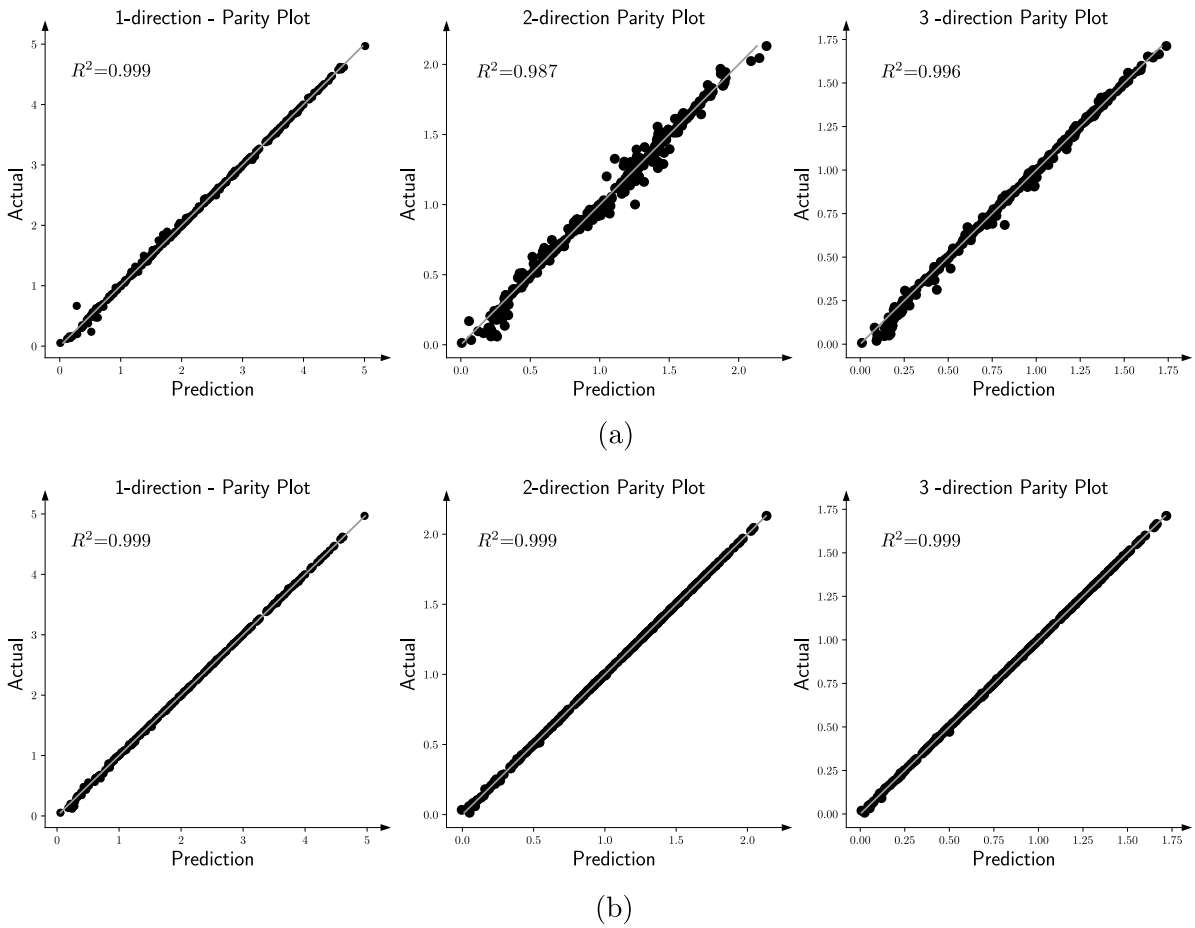

Fig. 7. Parity plots for comparison of predicted thermal conductivity vs actual conductivity in three principal directions when the GP is trained using (a) data set containing 50 points (b) data set containing 250 points. Both axes in units of W/m °C.

Table 2
Mean Squared Error for GP surrogates.

GP Model	MSE using data set D_{50} (e_1)	MSE using data set D_{250} (e_2)	$\frac{e_1}{e_2}$
K_1 surrogate	15.99×10^{-4}	3.25×10^{-4}	4.92
K_2 surrogate	30.63×10^{-4}	0.28×10^{-4}	109.39
K_3 surrogate	7.29×10^{-4}	0.14×10^{-4}	52.07

denote as D_{50}, consists of 50 training points, and the second data set, which is denoted as D_{250}, contains 250 training points. We generate both training data sets by sampling the inputs using the Latin Hypercube design [55] and running the micromechanics model for the corresponding inputs. Once the optimal hyperparameters of the model are estimated, we generate a test data set of size 300 points (from a different Latin Hypercube design) to evaluate the models trained on D_{50} and D_{250}. Fig. 7 shows the mean predictions of the GPs compared to the actual value generated using the micromechanics model. Fig. 7(a) shows the mean prediction of the GP compared to the actual value when the GP is trained using 50 training points. Note that the mean prediction deviates from the 45-degree line indicating that the mean predictions cannot be used as a reliable estimate of the true response. However, on increasing the size of the training data set to 250, we see that the mean predictions compare much better to the actual value as shown in Fig. 7(b) (see Table 2).

Using a GP presents multiple advantages. Apart from its excellent capabilities to act as a surrogate in low-dimensional settings, the closed-form solutions of the GP allow its seamless integration into modern probabilistic languages. The GP predictions are incorporated into the inference model in Eq. (10). In Eq. (10), each z_i can be sampled from a Normal distribution conditioned on input as:

$$z_i|\mathbf{x} \sim \mathcal{N}(\mu_{f_i}, \tau_{f_i}^2), \tag{32}$$

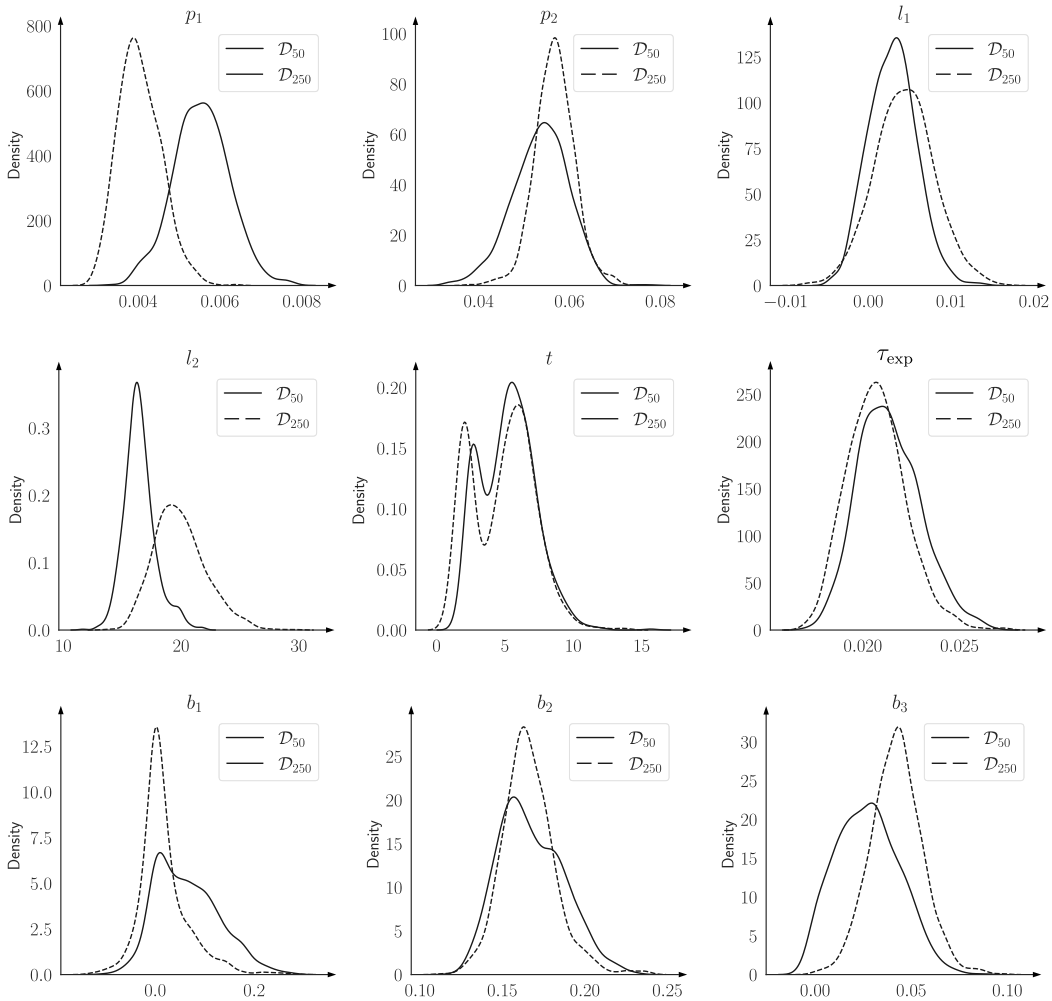

Fig. 8. Comparison of posteriors of all the parameters when different surrogates are used. $\mathcal{D}_{50}$ corresponds to the case for the inference where the GP was trained on 50 points, and $\mathcal{D}_{250}$ corresponds to the case for the inference where the GP was trained on 250 points.

where μ_i and τ_i^2 is evaluated using Eq. (29), and Eq. (30) respectively. Further, the analytical expressions for evaluating the mean and variance of the GP allow the use of gradient-based sampling techniques via automatic differentiation.

6. Results and discussion

Once the GPs are trained using the two datasets, we perform the Bayesian inference sampling using both GPs. We use the No-U-Turn sampler [56], as implemented in the Python package pymc3 [53]. The No-U-Turn sampler uses the gradient information of the posterior to sample efficiently in the parameter space. We draw a total of 100,000 samples to ensure the convergence of the Markov Chain to its stationary distribution. To remove any transient behavior (the period where the sampler finds the typical set in the joint distribution), we discard the first 1,000 samples. Out of the samples that remain, we retain one out of 40 samples to remove any chain autocorrelation. In this section, we will first discuss the inference results, and posterior predictive tests to ensure we identify the true posteriors of the parameters. The complete inference for 2 chains takes about 10 h using the GP trained on the dataset 250 on a standard PC with 16 GB RAM, an i7 processor at 3.4 GHz, and 4 cores. We anticipate significant speed up by using graphical processing units. We then proceed to use the inferred constituent properties, to predict the thermal conductivity of the composite in the LSAM system.

6.1. Inference results

We start by discussing the results for all the posteriors of the parameters in the inference model. We compare the inferred posteriors for two cases — first, the case where the GP was trained on dataset $\mathcal{D}_{50}$, and the second, the case where the GP was

trained on $\mathcal{D}_{250}$. The comparison of the posteriors can be seen in Fig. 8. There are several important observations to be made from Fig. 8. First, we observe that all the bias variables (i.e., b_i from the last row in Fig. 8) are more uncertain when the surrogate trained on $\mathcal{D}_{50}$ is used. This uncertainty in the posterior is induced by the epistemic uncertainty of the surrogate. This is an expected result indicating that an uncertain surrogate leads to uncertain posteriors, and this observation is consistent with the results in [47]. However, we notice that not all the parameters follow this trend. The other parameters, e.g., p_1 (first row of Fig. 8), clearly show a bias in the estimation. To clarify, the mean value of the posterior of p_1 is different when the surrogate is trained on different datasets. This points to an important aspect of the micromechanics model. The micromechanics model is only an approximation of the averaged properties of the composite's thermal conductivity and is not exact. The source of the error for the micromechanics model arises primarily from the dilute suspension assumption. In the model, invoking the dilute suspension assumption is crucial in the first step of homogenization (Mori–Tanaka model), which ignores fiber-to-fiber contact. However, in a composite, fiber-to-fiber contact is possible, and therefore it increases the effective thermal conductivity. Strictly speaking, the micromechanics model is an incorrect description of the true response. Hence, we observe different posterior distributions (with bias) when the different datasets are used to train the surrogate. Under these circumstances, an analyst must remove sources of controllable epistemic uncertainty (e.g. when building a surrogate). We achieve this by training the surrogate using 250 data points. To ensure that using the surrogate trained on $\mathcal{D}_{250}$ induced negligible epistemic uncertainty on the posteriors, we perform the inference using the mean value of the surrogate and compare it to the case where the entire predictive distribution of the surrogate is used. From Fig. 9, we see the mean predictions and the entire predictive distributions produce almost identical posteriors indicating that the effect of epistemic uncertainty is minimal. Therefore, for the remaining discussion, we choose to work with the posteriors obtained using the entire predictive distribution of the surrogate trained on $\mathcal{D}_{250}$. For completeness, we also trained a GP with dataset size of 500, and observed almost identical posteriors of the parameters, as shown in Fig. 9. Therefore, we could conclude that increasing the dataset size beyond 250 had no considerable effects on the inference. We have documented these findings in the Appendix. It is important to note that increasing the dataset size only reduces the epistemic uncertainty and does not remove the bias that could be induced by the micromechanics model. The assumption of dilute suspension does induce bias in the prediction that does not disappear with more data. The ideal way to deal with low-fidelity models would be to calibrate model errors as a GP, following [57]. Further, to make the best use of resources in building a surrogate, an analyst should consider using a sequential design of experiments [58] to query input locations till the epistemic uncertainty induced by the surrogate falls below a specified tolerance.

The posteriors of the constituent properties are plotted in Fig. 10. We observe that the credible intervals for the fiber thermal conductivities are wide. This is attributed to the fact that fiber thermal conductivity has minimal effect on the composite's thermal conductivity and therefore its inference becomes difficult. The consideration of fiber-to-fiber contact in a micromechanics model would change these results. From Fig. 10, we observe that there is an evident asymmetry in the posterior of the thermal conductivity in the transverse direction. This asymmetry is caused by the choice of the priors and insensitivity of the transverse fiber thermal conductivity to the composite thermal conductivity, given the micromechanics model, and is not a material behavior. To check the validity of the posteriors, we conduct a posterior predictive check, i.e., we check if the posteriors can indeed recreate the experimental data. This comparison is plotted in Fig. 11. From Fig. 11 we see that the predicted credible intervals cover most of the experimental data indicating a favorable data generating model.

6.2. Prediction of conductivity for a different microstructure

Up to this point, we have inferred the thermal conductivity of the constituent properties as a function of temperature. We have now set up everything we need to make thermal conductivity predictions on the LSAM system. To do so, we first infer the fiber orientation state of the material printed in LSAM using tensile tests conducted in the 1- and 3-directions. As discussed, this step presents the major advantage of our framework over conventional transfer learning frameworks. We identify that the change in composite thermal conductivity is brought about by a change in microstructure and infer microstructural information using tensile tests. Following [29], we obtain the fiber orientation state of the material printed in LSAM as $\mathbf{A}_2 = \mathrm{diag}(0.58, 0.32, 0.1)$. Next, the fiber length and therefore the fiber aspect ratio was measured as discussed in the Section 3. It is important to mention that the thermal conductivity of misaligned SFRPs depends highly on the fiber orientation state and fiber length as shown in [59]. Since we infer the fiber orientation and measure the fiber length, we do not conduct a sensitivity study as to how they can affect the thermal conductivity of the polymer. We wish to re-emphasize that the goal of our work is to present the fundamental framework of on how to transfer material properties across printing systems. To make thermal conductivity predictions of the composite on the LSAM system, we sample 500 points from the posterior of the inferred constituent properties (shown in Fig. 10) for 100 values of temperatures between 20 °C and 200 °C. We run the micromechanics model (not the surrogate) for the 500 by 100 matrix to obtain convergent statistics. Note that we do not use the inferred bias variables to estimate the thermal conductivity of the polymer. This is motivated by the fact that they were inferred for the fiber orientation state of the material printed in CAMRI. A change in the fiber orientation state would change the value of the bias (or model error) variables and therefore loses its validity prediction of thermal conductivity in the LSAM system. To validate our predictions we conduct extensive experimental measurements on samples extracted from panels printed on the LSAM system. We identify that the fiber orientation state varies within a printed bead of the LSAM system (the printed bead has a dimension of 20 mm in width and 5 mm in height). To capture the effect of fiber orientation variation, we carefully extract thermal conductivity specimens from various regions of the printed bead. We start by directing the readers' attention to Fig. 12. In the 1-direction (printing direction), we extract two specimens with different diameters. We extract one large sample and one small sample. The illustrations in Fig. 12 are to direct the reader as to where the samples are approximately extracted relative to the location of the beads. For the large sample, we anticipate that the fiber orientation closer to the edges of

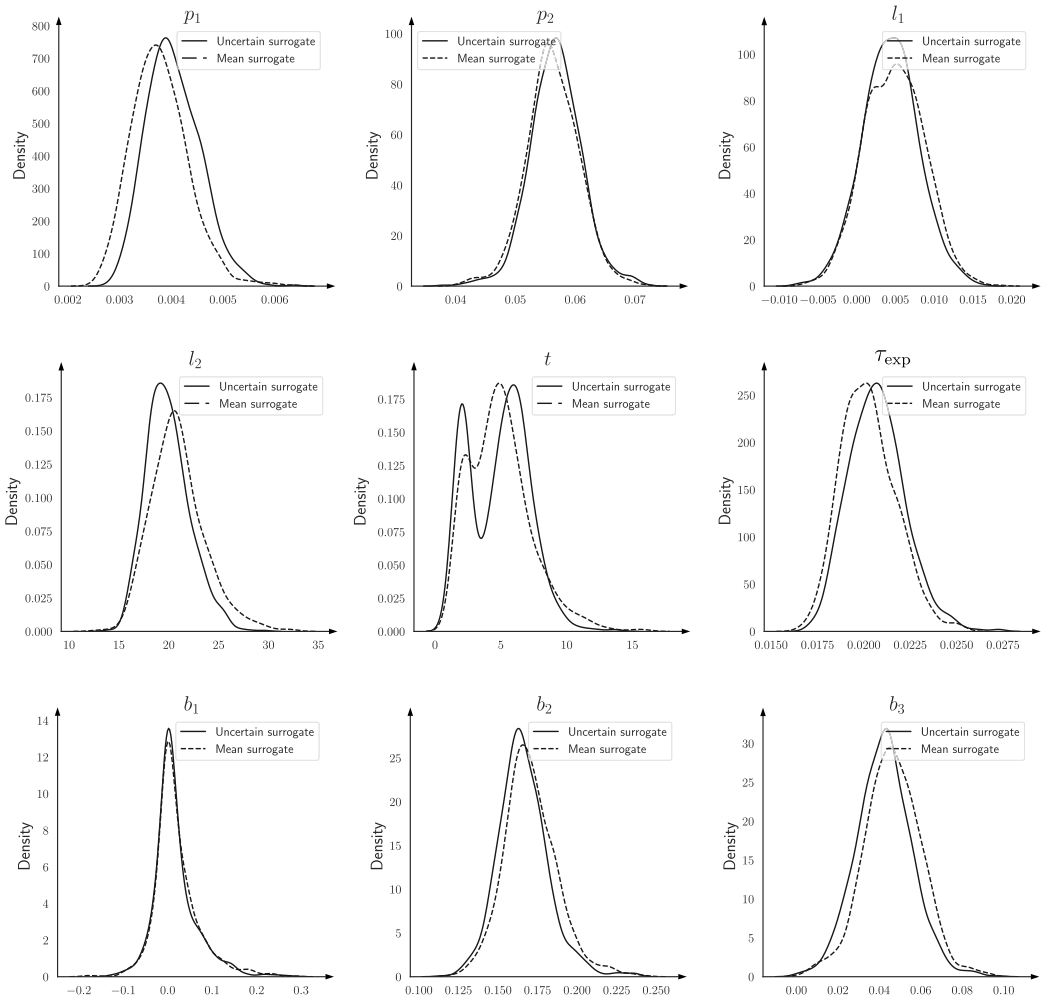

Fig. 9. Comparison of posteriors of all the parameters when the surrogate trained on $\mathcal{D}_{250}$ is used. Using the mean predictions and the entire predictive distributions (denoted as uncertain surrogate) produces almost identical posteriors indicating that the effect of epistemic uncertainty is minimal.

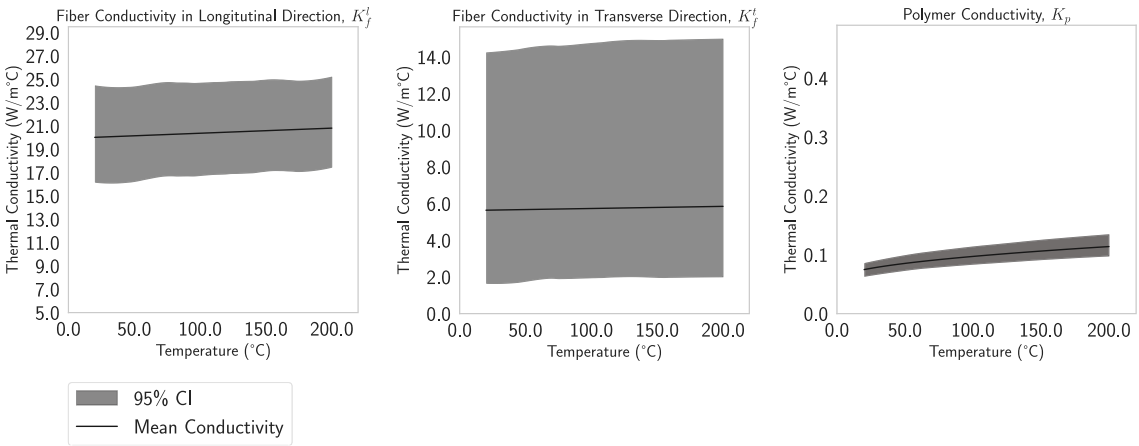

Fig. 10. Posterior distribution of the constituent properties as a function of temperature. From left to right, the thermal conductivity of the fiber in the longitudinal direction, the thermal conductivity of the fiber in the transverse direction, and the thermal conductivity of the polymer.

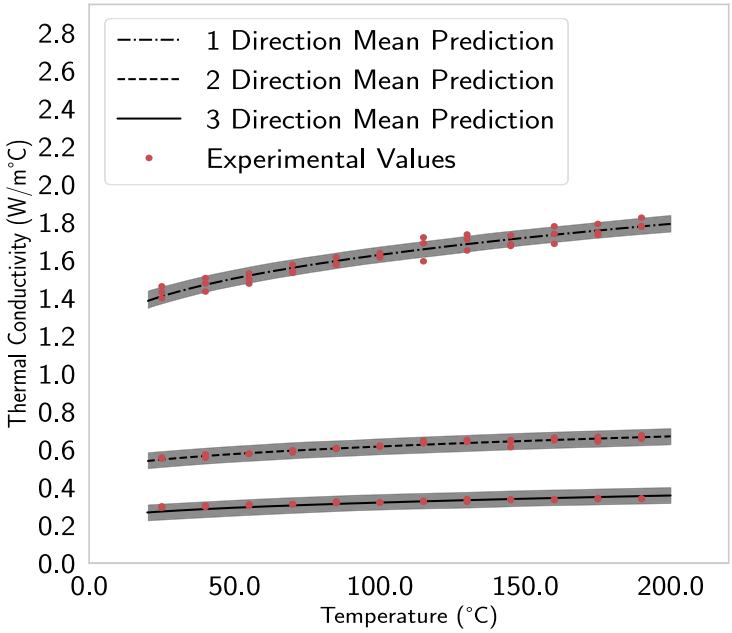

Fig. 11. Posterior predictive checks to confirm the validity of the inferred posteriors.

Table 3
Experimental Samples Dimensions for LSAM.

Direction	Sample type	Diameter (mm)	Thickness (mm)	Mass (mg)
1-Direction	Large Sample	25.21	3.93	2422.88
2-Direction	Edge Sample	25.49	5.58	3470.26
1-Direction	Small Sample	12.52	3.90	585.41
2-Direction	Center Sample	25.35	5.65	3500.92
3-Direction	Small Sample	12.60	3.97	592.75

the bead will contribute significantly to the composite thermal conductivity. Shear flow in the printing nozzle preferentially aligns the fibers in the printing direction, and we expect to see the effect of this on the thermal conductivity measurements. i.e., we expect a higher value in thermal conductivity. For the small sample extracted in the 1-direction (see Fig. 12) we expect the fibers to be less aligned in the 1 direction, as this region of the bead is away from the influence of the shear flow along the nozzle. Therefore, we expect a lower value in thermal conductivity. The dimensions of the large and small samples in the 1-direction are tabulated in Table 3. Next, we consider the extraction of samples in the 2-direction (the direction transverse to the printing direction). In this case, we extract two large samples at different locations along the width of the bead. Again, we motivate this in anticipation of fiber orientation variation within the bead. The locations where the samples are extracted in the 2 direction are illustrated in Fig. 12 and the respective sample dimensions are tabulated in 3. Finally, we extract one sample in the 3-direction as illustrated in Fig. 12. Only one sample is extracted since variation in fiber orientation in 3 direction is expected to be minimal. Once the samples are extracted, an important step is to decide a detection area for the thermal diffusivity step. The experiment is performed by shooting an energy pulse in a localized area on the surface of the sample and measuring the transient thermal history on the opposite face. Therefore, we choose a detection area that covers approximately an integer number of beads for the three principal directions. The detection area chosen relative to the sample sizes are shown in Fig. 13

We compare our thermal conductivity predictions to the experimentally measured thermal conductivities in Fig. 14. From Fig. 14 we see that our predictions agree well with one set of measurements. To clarify, from Fig. 14, we observe that our predictions in the 1-direction and 2-direction agree well with the experimental measurements highlighted in blue. The reason for this relates back to the step where the fiber orientation state was inferred using tensile tests.

Let us first focus on the measurements in 1-direction and notice that our predictions agree well with the thermal conductivity measurements made on the small samples (blue markers). While conducting the tensile tests, the edges of the bead are machined down to form a plane surface. In doing so, regions of high orientation in the printing direction (1-direction) are removed. Consequently, while inferring fiber orientation state from tensile tests, we lose some information from these regions of high alignment. Further, note that we under-predict the 1-direction thermal conductivity of the composite. This can be explained by the fact we do not consider fiber-to-fiber contact in the micromechanics model. The average value of a_{11} in the second order orientation tensor, $\mathbf{A}$, is lower in the material printed in LSAM as compared to the CAMRI. This, in turn, increases the possibility of fiber–fiber

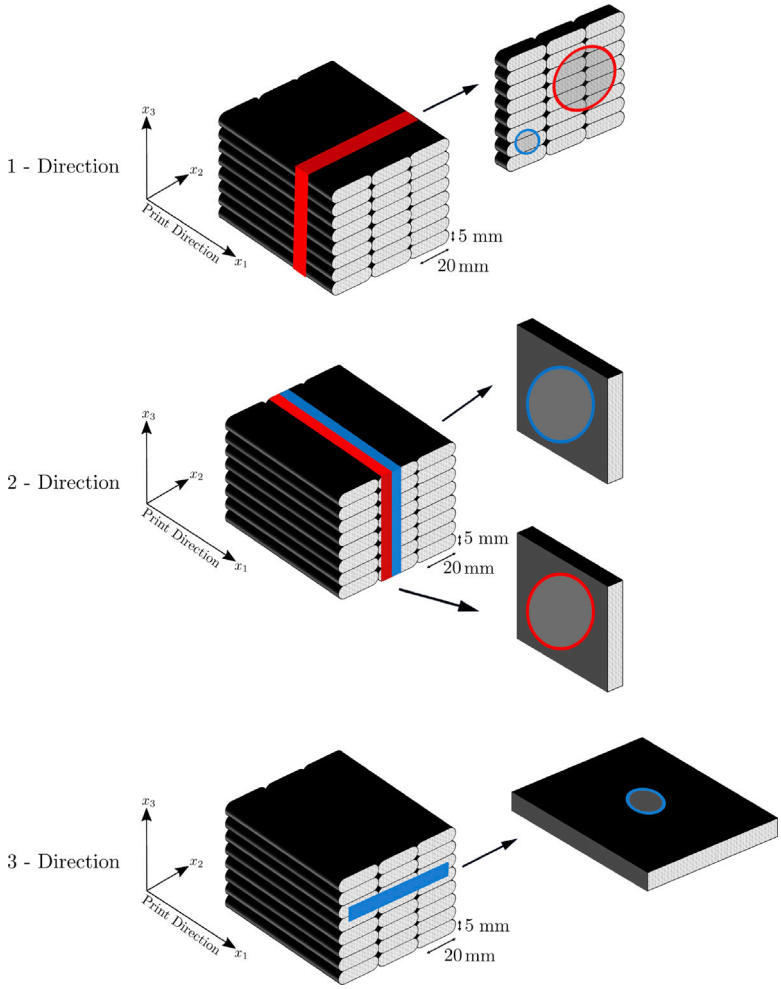

Fig. 12. Extracting experimental samples for LSAM in 1-Direction - small sample (red) and large sample (blue), 2-Direction - the center of the bead sample (blue) and the edge of the bead sample (red), and 3-Direction - small sample (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

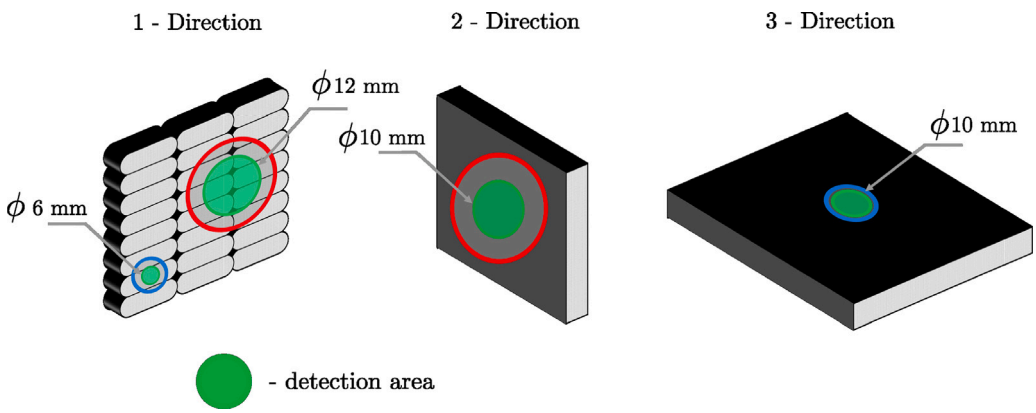

Fig. 13. Sample sizes and corresponding detection areas used in thermal diffusivity measurements.

interaction (since they are less aligned in the 1-direction) and enables paths of less resistance for heat transfer. Modeling the fiber-to-fiber contact is beyond the scope of this work since our focus is to develop a general framework that can utilize any suitable micromechanics model. Further, note that we observe a high value for thermal conductivity when measured using the large sample. While we anticipate that the larger sample might homogenize variations over the bead and therefore would follow our predictions closely, this is not the case. We point out that the laser flash technique does not shoot the energy pulse over the entire specimen. Rather it shoots the energy pule in a circular region that is half the size of the specimen (the large specimen is 1" in diameter and the detection area is 0.5"). This is done to reduce energy losses from the ends of the specimen that can bias the diffusivity measurements. Given the relative size of the detection area, we observe that the detection area covers regions around the ends of the bead (see Fig. 13). We expect shear forces to dominate and align the fibers along the printing direction towards the ends of the bead. This preferential alignment of the fibers increases the thermal conductivity of the composite and that is why we observe the high thermal conductivity of the composite using the large sample.

Moving to the 2-direction measurements in Fig. 14, we again observe that our thermal conductivity predictions closely follow the experimental measurements highlighted in blue. These experimental samples are extracted from the center of the bead along the width. The fiber orientation along the 2-direction is expected to be higher in this region (and consequently the thermal conductivity is higher) than the specimen extracted closer to the edge (highlighted using red). Since we machine the edges of the specimen before conducting tensile tests, we, in turn, measure only an average fiber orientation state, and therefore unable to predict the lower thermal conductivity in the 2-direction Finally, we observe from Fig. 14, that our thermal conductivity measurements in the 3-direction follow closely with the experimental measurements. The experimental results indicate a significant variation in fiber orientation within a printed bead of the LSAM system. We only infer an average fiber orientation from tensile tests as outlined in [29]. To predict the variation of thermal conductivity within a bead, one needs the fiber orientation variation within a bead. One possible way to extract the distribution of "average" fiber orientation is using the method discussed in [35]. However, recent work by Pibulchinda et al. [16] presents a promising approach to estimating fiber orientation variation accounting for anisotropic viscosity induced by the fiber orientation state. Using the information on fiber orientation variation within a bead, one could predict the entire field of thermal conductivity using our approach. It is important to point out that the significance of our work lies in the capability to predict material properties just using tensile tests, by making use of prior knowledge. The major impact of this work is a reduction in characterization times required to populate material properties required as inputs for manufacturing digital twins. As an example, we show that one can predict the thermal conductivity of the composite to an agreeable extent without repeating the experiments for the same material printed on a different processing system.

7. Validation of predicted thermal conductivities

From the predictions of the thermal conductivities using tensile tests, we observe that our predictions are in good agreement for samples extracted towards the center of the bead. However, the samples extracted towards the edges of the bead exhibit some discrepancies in thermal conductivity from the predicted values. To examine the effect of these discrepancies in the thermal history of a printed part, we conduct process simulations for a part printed on the LSAM system. We 'virtually' print a geometry of interest using two thermal conductivities, i.e., an effective experimental thermal conductivity and the mean predicted thermal conductivity. The effective experimental thermal conductivity is obtained by considering an average over the 2 specimens tested,(for the 1-direction and 2-direction) as conductors in series. To clarify, the effective experimental thermal conductivity measured at each temperature is evaluated using the formula:

$$K_{eff} = \frac{2K_a K_b}{(K_a + K_b)}, \tag{33}$$

where K_{eff} is the effective experimental thermal conductivity. For K_{eff} in 1-direction, K_a represents the thermal conductivity measured using the small sample, and K_b is the thermal conductivity measured using the large samples. For K_{eff} in 2-direction, K_a represents the thermal conductivity measured using the center sample, and K_b is the thermal conductivity measured using the edge samples. Note that we only consider this averaging for the 1-direction and 2-direction. We conduct process simulations only for the average response from the predicted thermal conductivity. The geometry we consider is a box structure with features to examine the effect of 1-direction, 2-direction, and 3-direction thermal conductivity. Fig. 15 shows the geometry considered for investigation. We name each section to identify them seamlessly. To study the effect of 1-direction thermal conductivity, we print a straight section in section B and this serves as a good indicator to examine the effects of thermal conductivity. Next, to study the effects of 2-direction thermal conductivity, we consider section A. Note the printing pattern in section A. Section A consists of three beads along the y-direction and is printed in a pattern such that 2-direction thermal conductivity plays an important role in the transient temperature history. From Fig. 15 we see the printing pattern in Step 2 leaves a gap in between two beads in section A. This gap is filled after an additional bead is laid down in section B. We will study the straight wall (in the z-direction) to study the effect of 3-direction thermal conductivity. Note, that this geometry is chosen only as a virtual experiment, and does not represent a real case scenario. This printing strategy allows us to examine the effect of the difference between the measured thermal conductivity and predicted thermal conductivity.

A voxel FE mesh is generated based on the event series of the geometry. Thus, elements are only present where the print head deposits material. Beads are discretized with two elements in the width and two in the height which produces elements with dimensions of 10.69 mm in width and 2.57 mm in height. Linear brick elements type DC3D8 are used. To simulate the material in the molten state prior to deposition, the temperature of the nodes in the FEA mesh is initialized to the melt temperature. The

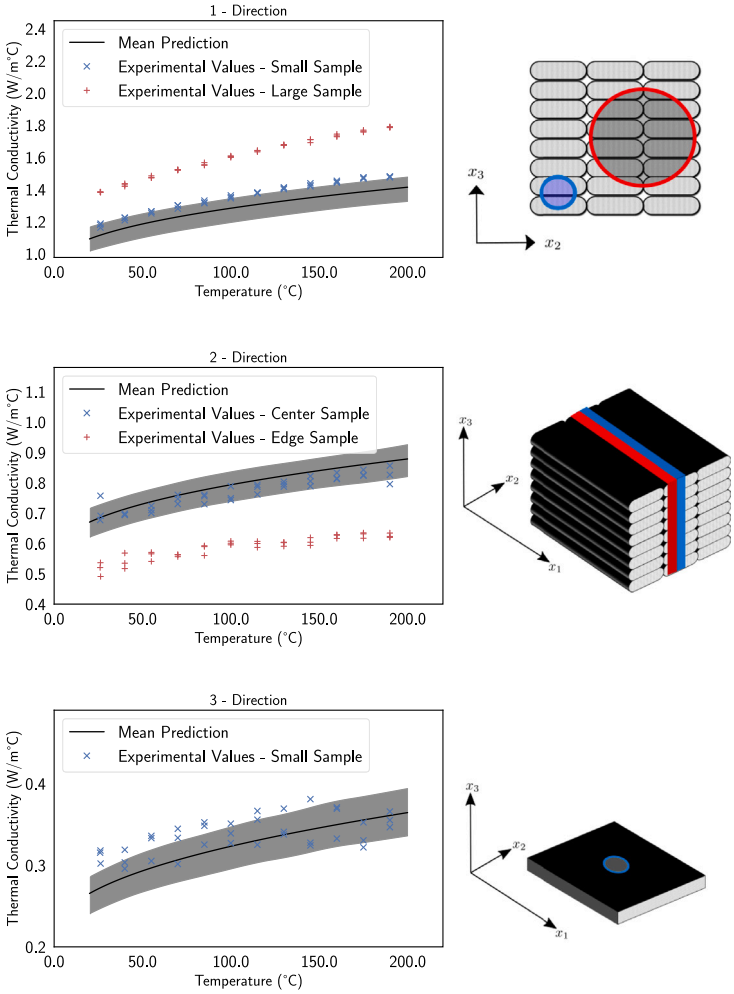

Fig. 14. Comparison of thermal conductivity predictions to experimentally measured values.

Table 4
Parameters of the process and material used in the simulation of the printing process of the plate.

Parameter	Value
Molten Temperature (°C)	365.0
Deposition Speed (mm/min)	2460.0
Ambient Temperature Convection (°C)	25.0
Ambient Temperature Radiation(°C)	25.0
Heat Removed By Compactor (W)	50.0
Emissivity	0.95

bottom surface is set to a zero heat flux boundary condition. The convection conditions developed at the surface of the geometry are governed by a correlation equation that provides a local convection coefficient. More details can be found in [4]. The LSAM system uses a roller to compact the material deposited and the heat removed by this compaction mechanism is considered in the analysis. The printing conditions simulated are summarized in Table 4.

We first consider a qualitative analysis of the temperature developed in the x-y plane. This serves as a qualitative check to understand the effect of using the effective experimental conductivity compared to the mean predicted conductivity. Fig. 16 shows the qualitative comparison in the plane and we observe that except for minor changes, the results compare extremely well. Next, we also qualitatively compare the temperature field in the stacking direction (z-direction) in Fig. 17. From Fig. 17 we see identical temperature fields when using the effective experimental thermal conductivity compared to using mean predicted thermal conductivity.

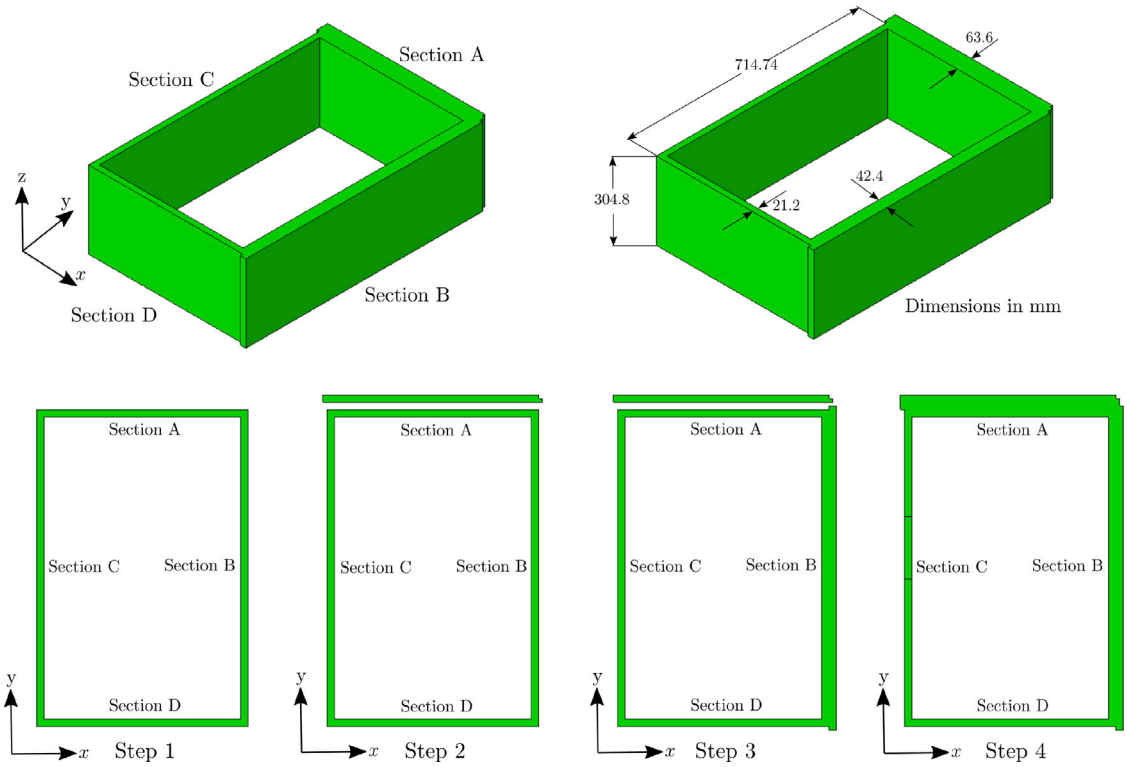

Fig. 15. Geometry considered and the printing scheme used to print.

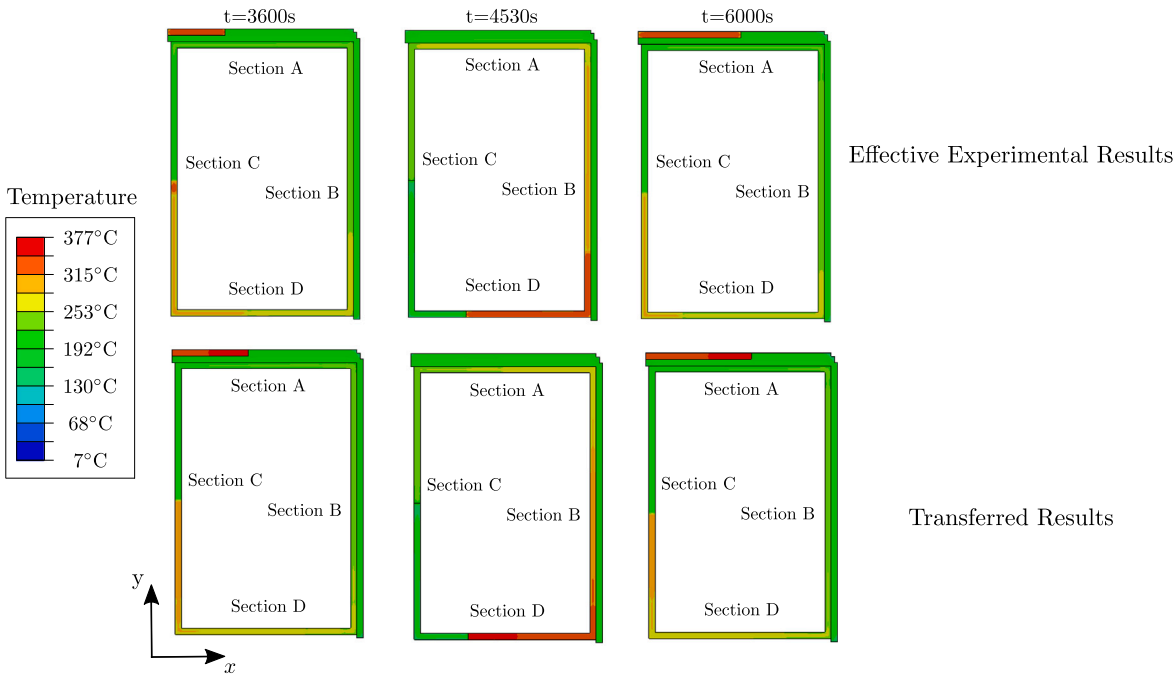

Fig. 16. In-plane temperature field comparisons for three-time increments during the printing process.

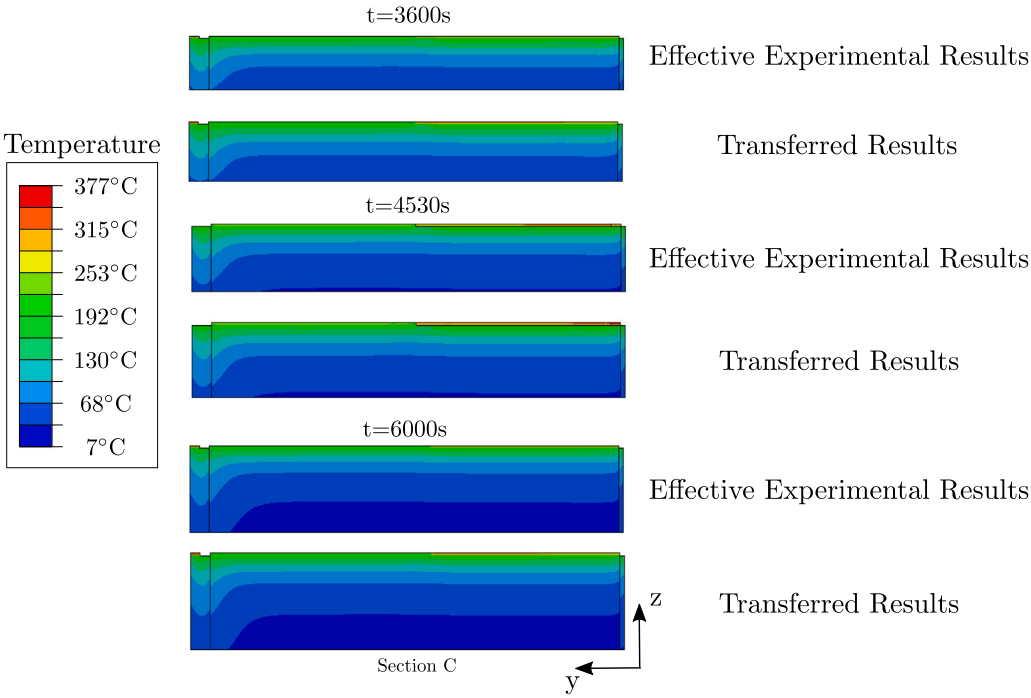

Fig. 17. Stacking direction temperature field comparisons for three-time increments during the printing process.

Next, we conduct a quantitative analysis to estimate the effect of using the mean predicted thermal conductivity. We aim to analyze the effect of each component of the thermal conductivity separately. First, we consider the effect of the 1-direction thermal conductivity. To do so, we query five points along the stacking direction from section B of the geometry (see Fig. 18(a)) in layers 10, 15, 23, 33, and 41 consecutively. The points are queried at this location since it is a straight section and is best suited to study the 1-direction thermal conductivity. We plot the transient heat transfer history of these points in Fig. 18(a). From Fig. 18(a) we observe that the transient temperature history using the mean predicted thermal conductivities is in excellent agreement with the temperature history using the effective experimental conductivity. We also plot the percentage error as a function of time for the various points queried in Fig. 18(a), as shown in the last row. From the error plots, we see that the percentage error is less than 4%. Further, note that there are some fluctuations in error as the elements get activated. We noticed in our simulations that the solver makes some cutbacks in the time increment used (set to 1) to obtain convergent results. However, the cutbacks made in the simulation using the mean predicted conductivity is different from the cutbacks made when using the effective experimental conductivity. Therefore, there is a small difference in the times the same element is activated (a difference of around 2s). Nonetheless, we choose to present our results as is. The error fluctuations might decrease if we choose a very small time increment.

Next, we consider the effect of the 2-direction thermal conductivity. To do so, we query five points along the stacking direction from section A of the geometry (see Fig. 18(b)) in layers 7, 27, 38, 50, and 57 consecutively. The points are queried at this location since it is a section with three beads and serves as a good indicator for conduction in the 2-direction. We plot the transient heat transfer history of these points in Fig. 18(b). From Fig. 18(b) we observe that the transient temperature history using the mean predicted thermal conductivities is in excellent agreement, as evident from the error plots in the last row with less than 6% error, with the temperature history using the effective experimental conductivity.

Finally, we consider the effect of the 3-direction thermal conductivity. Here, we query five points along the stacking direction from section C of the geometry (see Fig. 18(c)) in layers 13, 20, 31, 37, and 51 consecutively. The points are queried at this location since it is a straight wall where the effects of 3-direction thermal conductivity are pronounced. We plot the transient heat transfer history of these points in Fig. 18(c). From Fig. 18(c) we observe that the transient temperature history using the mean predicted thermal conductivities is in favorable agreement with the temperature history using the effective experimental conductivity. Again, the favorable agreement is evident from the error plots (in the last row of Fig. 18(c))showing less than 2% error.

These results show that an analyst can transfer material properties across extrusion systems effectively. Our framework presents considerable gains in characterization times since we predict the thermal conductivity was transferred to the LSAM system just using tensile tests. Further, the Bayesian formalism of the problem considers sources of uncertainty in the entire process which is crucial to develop probabilistic simulation frameworks for composites additive manufacturing.

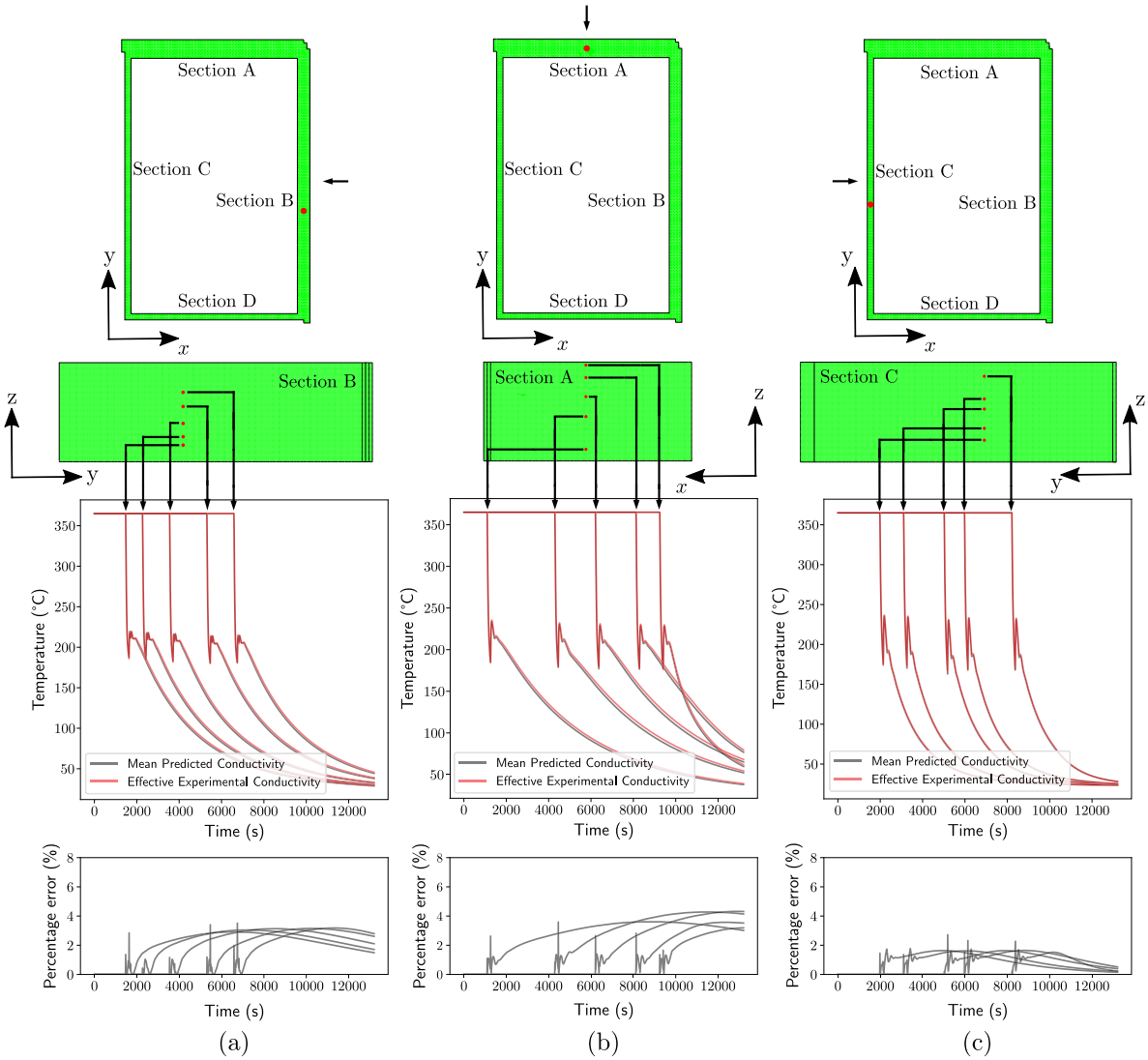

Fig. 18. Comparison of time–temperature history using mean predicted thermal and effective experimental conductivity. (a) points are queried in section B to study the effect of 1-direction thermal conductivity (b) points are queried in section A to study the effect of 2-direction thermal conductivity (c) points are queried in section C to study the effect of 3-direction thermal conductivity. The percentage errors a function of time is shown in the last row.

8. Conclusions

This work presented a framework to predict the thermal conductivity of a composite printed on the LSAM system by only conducting tensile tests. This is made possible by the availability of prior information on the thermal conductivity of the same material printed on the different system (CAMRI) and the ability to infer fiber orientation state from tensile tests [29,35]. We presented a rigorous statistical approach and outlined the effect of measurement uncertainty and surrogate uncertainty on inferring constituent properties. We developed the mathematical tools to account for these uncertainties and presented guidelines to build reliable surrogates that do not bias the inference of constituent properties. This study focused on the thermal conductivity of the composite, but the statistical framework developed is such that it can transfer other material properties, including coefficient of thermal expansion and viscoelastic properties.

This framework is of importance since it has the potential to reduce the extensive experimental characterization required to enable manufacturing simulations. An analyst could potentially predict the properties of material manufactured on a large system like the LSAM by conducting characterization effectively in a smaller system, thereby reducing material and lab light of the presented framework, we identify directions for future research. A natural extension of this work is to consider a micromechanics model that considers fiber-to-fiber contact. This could potentially improve our predictions and be more representative of the true material microstructure. Beyond, the scope of the current work, our framework is amenable to account for manufacturing and

material composition variations. Manufacturing variations manifest as microstructural changes when considering the extrusion of fiber reinforced polymers. We have shown in our previous work [35] that one can capture variations of fiber orientation in different regions of a printed part by conducting tensile tests. Once the variations of fiber orientation are inferred one could always estimate the thermal conductivity of the composite for all possible variations of the inferred fiber orientation state. Recent work by Pibulchinda et al. [16] presents a promising approach to estimating fiber orientation variation accounting for anisotropic viscosity induced by the fiber orientation state. Using the information on fiber orientation variation within a bead, one could predict the entire field of thermal conductivity using our approach. Further, our method is applicable is to transferring information across simpler printing systems as long as the change in properties are attributed to a change in microstructure.

As a future direction of research, an equivalent framework needs to be developed for other properties, including the anisotropic coefficient of thermal expansion. The anisotropic coefficient of thermal expansion is an important property that governs the residual stresses that develop and, in turn, controls the deformation at the end of the printing process. This development needs to be validated on complex real-world geometries. Further, we are working on improving the computational time for the complete workflow by utilizing graphical processing units to speed up the MCMC sampling.

CRediT authorship contribution statement

Akshay J. Thomas: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Mateusz Jaszczuk:** Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft. **Eduardo Barocio:** Conceptualization, Data curation, Funding acquisition, Supervision, Writing – review & editing. **Gourab Ghosh:** Software. **Ilias Bilionis:** Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing. **R. Byron Pipes:** Funding acquisition, Investigation, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. Effect of GP trained using various datasets on the inference of constituent thermal conductivities

In the results section we compared the inferred posteriors for two cases — first, the case where the GP was trained on dataset $\mathcal{D}_{50}$, and the second, the case where the GP was trained on $\mathcal{D}_{250}$. To further validate that using a dataset of size greater than 250 does not reduce the epistemic uncertainty, we consider a dataset of size of 500 (called $\mathcal{D}_{500}$) to train the GP. The posteriors of the parameters of the inference model are plotted in Fig. A.1, using the three GPs. We observe from Fig. A.1 that the posteriors induced by using the GP trained on $\mathcal{D}_{250}$ and $\mathcal{D}_{500}$ are almost identical, indicating that there is no considerable reduction in epistemic uncertainty by increasing the training dataset of the GP from 250 to 500. The only significant change in the posterior is in variable t. However, we further plot the posterior of the constituent thermal conductivities using the results obtained when using the GP trained on $\mathcal{D}_{500}$ in Fig. A.2. Note that the posterior of the constituents when using the GP trained on $\mathcal{D}_{500}$ (Fig. A.2) is nearly identical to the posterior of the constituents when using the GP trained on $\mathcal{D}_{250}$ (Fig. 10).

Therefore, we conclude that increasing the dataset for training the GP beyond $\mathcal{D}_{250}$ has no significant advantage in terms of reduction of epistemic uncertainty.

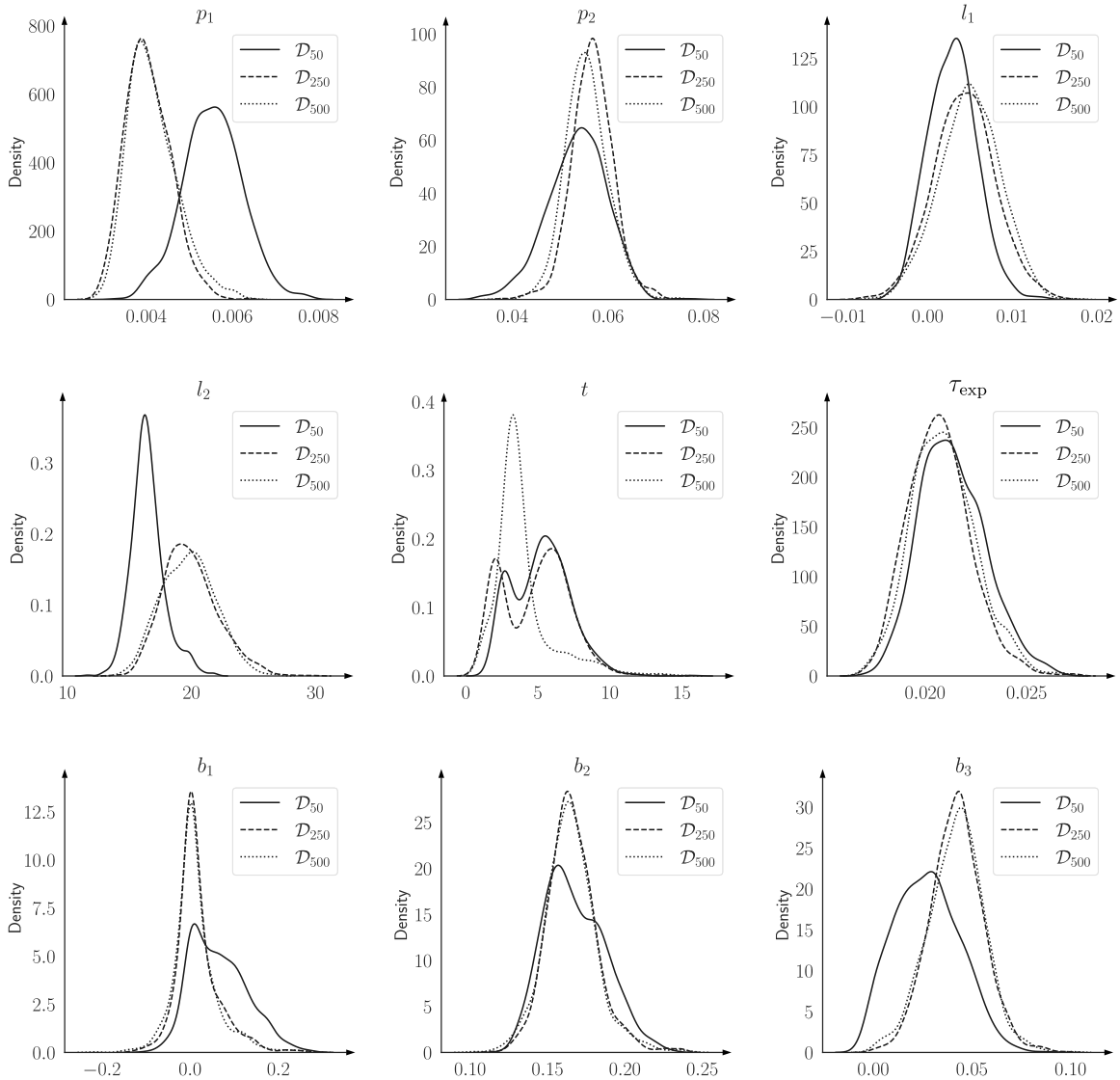

Fig. A.1. omparison of posteriors of all the parameters when different surrogates are used. D_{50} corresponds to the case for the inference where the GP was trained on 50 points, D_{250} corresponds to the case for the inference where the GP was trained on 250 points, and D_{500} corresponds to the case for the inference where the GP was trained on 500 points.

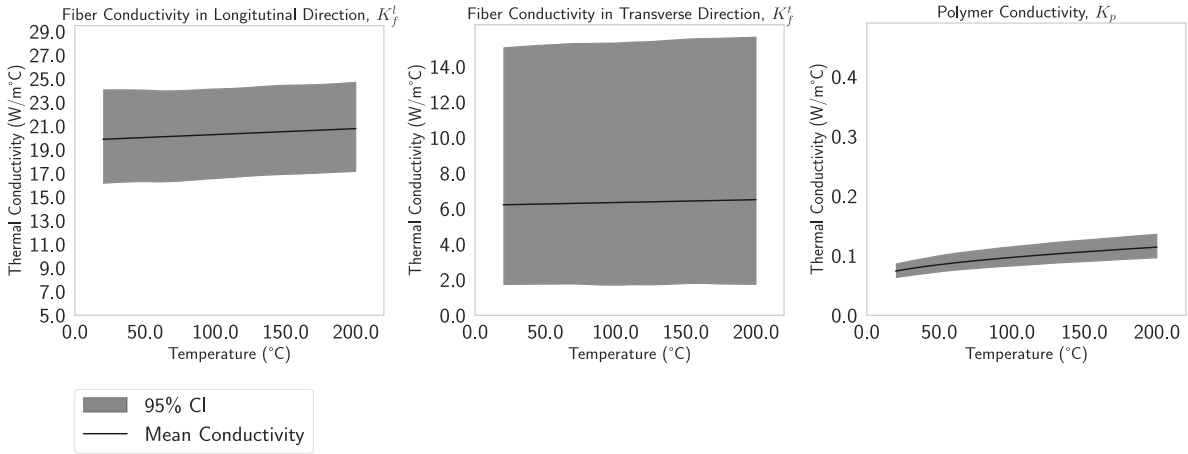

Fig. A.2. Posterior distribution of the constituent properties as a function of temperature. From left to right, the thermal conductivity of the fiber in the longitudinal direction, the thermal conductivity of the fiber in the transverse direction, and the thermal conductivity of the polymer, when the GP trained on $\mathcal{D}_{500}$ is used.

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